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PPG-Based Atrial Fibrillation Detection with Signal Quality Assessment and Edge-AI Deployment

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Abstract

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia and is associated with increased risk of stroke, heart failure, and mortality [1, 2]. Electrocardiography (ECG) remains the clinical diagnostic standard, but intermittent ECG can miss asymptomatic or paroxysmal episodes [1–3]. Photoplethysmography (PPG) offers a scalable route for continuous rhythm screening, but it is an indirect optical measurement of peripheral blood volume and is sensitive to motion artefacts, low perfusion, poor contact, and non-AF irregular rhythms [4–6].

This project develops a PPG-only AF screening pipeline combining preprocessing, signal quality assessment (SQI), hybrid deep learning, record-level aggregation, and embedded deployment feasibility work. PPG is segmented into 30-second windows, band-pass filtered, z-score normalised, and converted into pulse timing, morphology, rhythm, and quality features. The offline classifier, RhythmMorphologyFusionNet, combines time-domain waveform morphology, log-magnitude STFT evidence, and a 17-feature rhythm/quality vector using gated fusion. Unreliable windows are rejected using an SQI gate, and segment probabilities are aggregated at record-event level using quality-weighted averaging.

MIMIC PERform AF is used as a small controlled baseline [7, 8], while MIMIC-III-Ext-PPG v1.1.0 prefix groups p00–p02 are used as the main large-scale experiment [9, 10]. The p00–p02 AF-versus-sinus-rhythm subset contains 789,864 SQI-accepted segments, including 101,796 AF and 688,068 SR segments. Training uses metadata folds 0–7, validation uses fold 8, and testing uses fold 9.

MIMIC PERform AF is reported only as a controlled sanity check, where the model achieved perfect record-level test metrics on five held-out records and segment-level F1 of 0.9558. On MIMIC-III-Ext-PPG p00–p02, the final SQI pipeline achieved record-level test accuracy 0.9536, sensitivity 0.8241, specificity 0.9729, precision 0.8194, F1 0.8217, AUROC 0.9272, and AUPRC 0.8365. Segment-level precision was lower at 0.6841, supporting record-level aggregation rather than single-window alerts. A compact 17-feature Nordic Edge AI model was also integrated into Zephyr firmware as a smoke-tested deployment-path demonstration, not as a reproduction of the full offline hybrid model. Overall, the results provide retrospective feasibility evidence for SQI-aware record-level PPG AF screening on p00–p02, while requiring full-prefix, multi-rhythm, hardware, and prospective validation before clinical use.

Contents

Acknowledgements	i
Abstract	ii
1 Introduction	1
1.1 Overview	1
1.2 Motivation	2
1.3 Contributions	2
1.4 Report Structure	3
1.5 Problem Framing and Success Criteria	3
1.6 Clinical and Engineering Hypotheses	4
1.7 Why the Project Is an Engineering System Rather Than Only a Classifier	4
2 Technical Background and Related Work	5
2.1 Atrial Fibrillation and the ECG Reference Standard	5
2.2 PPG as an Indirect Rhythm Signal	6
2.3 Digital Signal Representation and Preprocessing	6
2.4 Pulse Peaks and Inter-Beat Interval Features	7
2.5 Morphology and Signal Quality Features	8
2.6 Frequency-Domain and STFT Features	8
2.7 Machine Learning and Deep Learning Approaches for PPG AF Detection	9
2.8 Hybrid Fusion and Record-Level Screening Logic	9
2.9 Design Requirements for This Project	10
2.10 Critical Positioning Against Prior PPG AF Work	10
2.11 Why Signal Quality Is a First-Class Model Component	11
2.12 Screening Versus Diagnosis in Clinical Guidance	11
2.13 Evidence Map for the Thesis Contribution	11
3 Datasets	13
3.1 MIMIC PERform AF: Controlled Baseline	13
3.2 MIMIC-III-Ext-PPG v1.1.0: Main Dataset	13
3.3 Scope Used in This Project	14
4 Signal Processing and Feature Extraction	15
4.1 Segmentation	15
4.2 Filtering and Normalisation	15
4.3 Peak Detection	15
4.4 SQI Acceptance Criteria	16
4.5 Hand-Crafted Feature Vector	16
5 Deep Learning Model	18
5.1 Problem Formulation	18
5.2 Model Selection Rationale and Mathematical Background	18
5.3 RhythmMorphologyFusionNet Overview	19
5.4 Time-Domain Branch	20
5.5 Spectral Branch	20

5.6	Feature Branch	20
5.7	Gated Fusion and Classifier Head	21
5.8	Training Objective	21
5.9	Class Imbalance Handling	21
5.10	Data Augmentation and Test-Time Augmentation	22
5.11	Record-Level Aggregation	22
5.12	Threshold Selection	22
5.13	Dataset Provenance and Label Validity	23
5.14	Defined Scope of the p00–p02 Experiment	23
5.15	Data Leakage and Evaluation Discipline	23
5.16	Representativeness and Domain Shift	24
5.17	Rationale for the Three-Branch Architecture	24
5.18	SQI as Abstention Rather Than Filtering Only	24
5.19	Why Focal Loss and Class-Aware Sampling Are Used	25
5.20	Methodological Alternatives Considered	25
5.21	Offline–Embedded Model Separation	25
6	Evaluation Protocol	27
6.1	Primary and Secondary Evaluation Levels	27
6.2	Metrics	27
6.3	Threshold Selection and Operating-Point Interpretation	27
6.4	Segment-Level, Record-Level, and Episode-Level Reasoning	28
6.5	False Positives, False Negatives, and Screening Risk	28
6.6	Coverage and No-Call Behaviour	28
6.7	Reproducibility Artifacts	28
7	Results	29
7.1	Controlled Baseline: MIMIC PERform AF	29
7.2	Large-Scale Experiment Comparison	29
7.3	Feature-Level SQI Ablation	30
7.4	Neural Branch Ablation	30
7.5	Final SQI Validation and Test Results	31
7.6	Confusion Matrices	33
7.7	Threshold Analysis	33
7.8	Aggregation Ablation	34
7.9	Record-Level Error and Reliability Analysis	34
7.10	Operating-Point Interpretation	36
7.11	Calibration and Probability Meaning	36
7.12	Statistical Uncertainty and Dependence	36
7.13	Why Segment-Level Metrics Remain Secondary	36
7.14	Suggested Additional Evaluation Table	37
7.15	Integrated Interpretation of the Final Test Result	37
7.16	What the Ablations Do and Do Not Prove	37
7.17	Failure Modes Suggested by the Error Analysis	38
7.18	Comparison With Wearable Screening Studies	38
8	Embedded Edge AI Deployment	39
8.1	Deployment Rationale and Scope	39
8.2	Firmware Target	39
8.3	Nordic Edge AI Lab Model	40
8.4	Embedded Smoke Test	40
8.5	Deployment Result and Interpretation	40
8.6	Deployment Gap Analysis	41
8.7	Production-Oriented Verification Plan	41
8.8	Why Edge Inference Is Relevant for PPG Screening	42
9	Discussion	43
9.1	Main Finding	43
9.2	Segment-Level and Record-Level Metrics Tell Different Stories	43

9.3	Interpretation of Precision and Specificity	43
9.4	Role of SQI	43
9.5	Edge Deployment Interpretation	44
9.6	Ablation and Remaining Error Analysis Gaps	44
9.7	Clinical Workflow Interpretation	45
9.8	Why Precision Matters in This Dataset	45
9.9	Interpretation of the Validation–Test Gap	45
9.10	Contribution Relative to an MEng Project	45
10	Limitations	46
10.1	Human-Factors Risks in AF Screening	47
10.2	Bias, Coverage, and Unequal Abstention	47
10.3	Data Minimisation and Edge Processing	47
10.4	Regulatory Boundary	47
11	Future Work	48
12	Conclusion	49
A	Supplementary Technical Notes	50
A.1	Pipeline Rationale	50
A.2	Components Considered but Not Retained as Main Results	50
A.3	Offline and Embedded Model Separation	51
A.4	Reproducibility Notes	51
B	Ethical Considerations and Data Governance	52
B.1	Screening Versus Diagnosis	52
B.2	False Positive Burden	52
B.3	False Negative Risk	53
B.4	Coverage and Abstention	53
B.5	Bias and Subgroup Robustness	53
B.6	Data Governance	53

Chapter 1

Introduction

1.1 Overview

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia and is a major risk factor for ischaemic stroke [1, 2]. It is associated with increased risk of heart failure, cognitive decline, and mortality, and many AF-related strokes are preventable when the arrhythmia is detected early enough for follow-up and treatment [1, 2]. AF is typically defined from electrocardiographic evidence of irregularly irregular ventricular response without consistent discernible P-waves over clinically significant rhythm episodes [1–3]. The practical difficulty is that AF may be asymptomatic or paroxysmal, so short clinical ECG recordings and intermittent ambulatory monitors can miss infrequent episodes [1, 2].

Electrocardiography (ECG) remains the diagnostic standard because it directly records cardiac electrical activity and allows clinicians to inspect rhythm morphology [1–3]. However, ECG-based monitoring is not always convenient for continuous long-term screening. Wearable and continuous monitoring devices therefore provide an appealing opportunity to screen for rhythm abnormalities outside the clinic. Photoplethysmography (PPG) is especially attractive because optical pulse sensors are already present in many smartwatches, fitness trackers, and other low-burden devices. Each cardiac contraction produces a peripheral pulse wave, allowing PPG to capture heart-rate and rhythm information without electrodes or active user input during routine passive monitoring.

Large wearable studies have shown that PPG-based irregular-rhythm notifications can achieve high positive predictive value when alerts are issued conservatively [11–13]. At the same time, these studies reveal the central trade-off in PPG screening: to avoid false alarms, they often prioritise specificity by rejecting noisy data and requiring sustained irregularity, which can reduce sensitivity for shorter or lower-quality episodes. PPG is also an indirect measure of rhythm [4, 6]. Unlike ECG, it does not directly show atrial electrical activity, P-waves, PR intervals, or QRS morphology [1–4]. Instead, AF must be inferred from irregular pulse timing, beat-to-beat variability, and pulse morphology. These cues can be informative, but they can also be distorted by motion artefact, poor sensor contact, low perfusion, vasoconstriction, skin and device effects, and ambient light leakage [4–6, 14, 15]. Non-AF irregular rhythms such as premature atrial contractions, premature ventricular contractions, atrial flutter with variable block, or bigeminy-like patterns can also create irregular pulse intervals that resemble AF in PPG [5, 15].

This project addresses that challenge by studying a PPG-based AF screening pipeline built around explicit signal-quality handling, hybrid deep learning, record-level aggregation, and a compact deployment path for embedded inference. The central research question is:

Can a PPG-only AF screening pipeline achieve useful record-level AF detection on a large clinical PPG subset while remaining compatible with eventual embedded deployment?

The work is organised around three more specific research questions:

RQ1. Can SQI-aware PPG processing achieve useful retrospective record-event AF-versus-SR screening performance on MIMIC-III-Ext-PPG p00–p02?

RQ2. Does record-level aggregation improve practical screening behaviour compared with isolated 30-second segment predictions?

RQ3. Can the project’s 17-feature representation be connected to a feasible embedded inference path?

This work is framed as screening rather than diagnosis. A positive PPG prediction should indicate likely AF and motivate confirmatory assessment, but it should not replace diagnostic ECG [1–3]. The aim is therefore not to claim ECG-equivalent diagnosis from PPG, but to build and evaluate a robust, signal-quality-aware screening pipeline that can reduce false positives while preserving useful AF sensitivity.

1.2 Motivation

The motivation for this project is both clinical and engineering. Clinically, low-burden wearable screening could help identify AF that would otherwise remain undetected between clinic visits [1–3, 11–13]. Engineering-wise, wearable deployment imposes two competing demands: the pipeline must be accurate enough to be useful, but compact and reliable enough to be practical under noise, limited resources, and repeated real-world use.

This project focuses on PPG because it is already widely available in wearable hardware and can be collected passively [4, 6, 11, 13]. At the same time, that choice makes the problem harder. PPG is more vulnerable to artefact than ECG, and an AF detector that performs well on curated clean segments may still fail in realistic conditions [4, 5, 14, 15]. The project therefore emphasises preprocessing, peak reliability, signal quality assessment (SQI), and record-level evidence aggregation rather than treating the task as a waveform classifier alone.

The project also explores deployment feasibility from the start rather than only reporting offline model accuracy. This is included because the intended application is not a cloud-only analysis service, but a wearable or near-sensor monitoring workflow in which rhythm evidence is collected at the physical source. Moving at least part of the inference pipeline closer to the sensor can reduce dependence on continuous connectivity, avoid streaming all raw physiological data to a remote server, and support lower-latency screening or no-call decisions [16–18]. These considerations are especially relevant for patient-monitoring devices, where the hardware may be a specific-purpose healthcare device rather than a general consumer smartwatch.

The deployment component is therefore scoped as an engineering feasibility question. The main hybrid model uses waveform, spectral, and feature evidence, while a separate compact 17-feature model is integrated into Zephyr firmware as an embedded inference demonstration. This does not prove that the full research model has been deployed on-device. Instead, it tests whether the feature representation and inference workflow can be connected to the type of constrained embedded environment that would be required for a future wearable system.

1.3 Contributions

The main contributions are:

1. A staged experimental design using MIMIC PERform AF as a controlled baseline and MIMIC-III-Ext-PPG p00–p02 as the main large-scale benchmark.
2. A 30-second PPG preprocessing pipeline with band-pass filtering, peak detection, rhythm feature extraction, and explicit SQI-based segment filtering.
3. Implementation and evaluation of a hybrid deep learning model that combines raw waveform morphology, spectral content, and 17 hand-crafted rhythm/quality features, while explicitly testing whether the branches add stable value through neural branch ablation.
4. A record-level aggregation strategy that converts segment probabilities into episode-level predictions using quality-weighted averaging, with an aggregation ablation against mean, median, maximum, and top-k pooling.

5. Additional validation evidence including SQI coverage analysis, feature-level SQI ablation, neural branch ablation, a feature-only logistic baseline, representative waveform error review, and bootstrap confidence intervals for the main record-level metrics.
6. A Nordic Edge AI deployment path using a compact 17-feature model integrated into an nRF9160/Zephyr firmware project with an embedded smoke-test harness, presented as feasibility evidence rather than as a reproduction of the full hybrid research model.

The datasets are intentionally reported separately. MIMIC PERform AF is used as a controlled baseline, while MIMIC-III-Ext-PPG is used as the main large-scale evaluation. The results are not presented as a single mixed-dataset training result because the datasets differ in scale, label provenance, patient context, and acquisition conditions. Combining them without a careful domain-adaptation study would make interpretation less clear rather than stronger.

Table 1.1 summarises where each research question is answered in the report.

Table 1.1: Research questions and the main evidence used to answer them.

Research question	Main evidence
RQ1: SQI-aware AF-versus-SR screening	Chapters 3–7: dataset definition, SQI coverage, final hybrid-model metrics, bootstrap confidence intervals, and record-level error analysis.
RQ2: Record-level aggregation	Chapters 6–7: segment-level versus record-level results and aggregation ablation.
RQ3: Embedded inference path	Chapter 8: compact 17-feature Nordic Edge AI firmware integration and smoke-test harness.

1.4 Report Structure

The remainder of this report is organised as follows. Chapter 2 reviews the technical background and related work, focusing on the ECG reference standard, PPG signal characteristics, prior AF detection approaches, and the rationale for explicit signal-quality handling. Chapter 3 introduces the datasets used in the project and clarifies why MIMIC PERform AF and MIMIC-III-Ext-PPG are treated separately. Chapter 4 presents the signal processing and feature extraction pipeline, including segmentation, filtering, peak detection, and SQI criteria. Chapter 5 describes the implemented hybrid model, training objective, and record-level aggregation strategy. Chapter 6 defines the evaluation protocol and metrics. Chapter 7 presents the experimental results, including record-level and segment-level analysis, threshold behaviour, and error analysis. Chapter 8 describes the embedded Edge AI integration. Chapters 9 to 12 discuss the results, limitations, future work, and final conclusions. Appendix A provides supplementary technical notes, and Appendix B covers ethical considerations and data governance.

1.5 Problem Framing and Success Criteria

The central difficulty in this project is that the target clinical phenomenon and the available measurement modality are not the same object. AF is an electrical rhythm disorder, whereas PPG is a peripheral optical pulse measurement. A successful PPG model therefore cannot be assessed only by whether it separates two labels in a retrospective dataset. It must also show that its decisions are made under plausible signal conditions, that uncertain or corrupted windows can be rejected, and that isolated noisy segments are not over-interpreted as clinical events.

For this reason, the success criteria for this project are deliberately broader than segment-level accuracy. The pipeline is considered useful only if it satisfies four conditions. First, it must preserve a clear screening interpretation: a positive output indicates AF-like pulse evidence and should motivate ECG confirmation, not replace ECG diagnosis. Second, it must include an explicit reliability mechanism so that poor-quality windows can be excluded or down-weighted. Third, it must convert multiple 30-second windows into a record-event decision, because a practical alert should be supported by repeated evidence rather than one local probability. Fourth, it must show

at least a credible route toward embedded inference, because a PPG screening system is ultimately tied to the sensing hardware rather than to an unconstrained offline environment.

This framing also explains why this thesis does not optimise for a single metric. Accuracy alone is insufficient under class imbalance: a model could achieve high accuracy by favouring the majority SR class. Sensitivity alone is also insufficient because excessive false positives would create unnecessary alerts and reduce trust. Precision and specificity are therefore important for an alerting system, while AUROC and AUPRC provide threshold-independent evidence that the model ranks AF and SR record-events differently. The final threshold is selected by validation record-level F1 because the intended output is a record-event screening decision and because F1 balances AF detection against positive-alert reliability.

1.6 Clinical and Engineering Hypotheses

The implementation can be viewed as testing three linked hypotheses. The first hypothesis is that PPG-derived rhythm information is useful for retrospective AF-versus-SR screening when unreliable windows are removed. This is tested by the SQI-accepted p00–p02 evaluation and by comparing segment-level and record-level behaviour. The second hypothesis is that repeated evidence is more reliable than isolated window evidence. This is tested by record-level aggregation and by the aggregation ablation against median, maximum, and top-k pooling. The third hypothesis is that a feature representation based on rhythm and quality variables can support an embedded inference path. This is tested by the compact 17-feature Nordic Edge AI model, while keeping it separate from the main offline hybrid model.

These hypotheses are intentionally modest. They do not claim that PPG can diagnose AF independently of ECG, that the selected SQI thresholds are universally optimal, or that the full hybrid model has been deployed on the microcontroller. Instead, they define a defensible MEng engineering contribution: a signal-quality-aware screening pipeline with quantified retrospective performance, clear operating-point analysis, and a concrete embedded proof of concept.

1.7 Why the Project Is an Engineering System Rather Than Only a Classifier

A simple classifier would take a waveform segment and output AF or SR. The implemented system is broader than this. It includes signal conditioning, pulse-peak detection, SQI rejection, feature extraction, waveform and spectral learning, probability calibration analysis, record-event aggregation, and embedded inference testing. This matters because many failures in PPG-based rhythm analysis occur before the classifier stage. A missed peak, duplicate peak, low-perfusion segment, or motion-corrupted waveform can create an apparently irregular pulse train even when the underlying rhythm is not AF. Treating the problem as a system therefore reduces the risk that the final model simply learns artefact patterns.

This system perspective is also useful for interpreting negative evidence. The branch ablation shows that the waveform-only neural variant was strong in one retraining run, and the feature-level SQI ablation shows only modest gains from some SQI components. These results do not undermine the system framing. Instead, they show that the pipeline contains several interacting components whose value is partly operating-point dependent. The safest conclusion is that learned PPG morphology, SQI-aware preprocessing, and record-level aggregation are all important parts of a practical screening workflow, while the exact fusion architecture still requires repeated-seed and broader validation.

Chapter 2

Technical Background and Related Work

This chapter gives the technical background needed for the implemented system. It first explains why AF is usually defined using ECG, then describes how PPG can provide an indirect rhythm signal. It then introduces the signal-processing, feature-extraction, signal-quality, and learning concepts used by the final pipeline. The purpose is not to survey every possible AF classifier, but to justify the specific design choices used in this project: 30-second PPG windows, peak-derived rhythm features, spectral analysis, explicit signal quality assessment, hybrid deep learning, and record-level aggregation.

2.1 Atrial Fibrillation and the ECG Reference Standard

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia. It is characterised by disorganised atrial electrical activity and an irregularly irregular ventricular response. Clinically, AF matters because it is associated with increased risk of ischaemic stroke, heart failure, cognitive decline, and mortality [1, 2]. Early detection is valuable because many AF-related strokes are preventable when the arrhythmia is identified and followed up appropriately [1, 2].

The difficulty is that AF is not always continuously present. Paroxysmal AF may begin and stop spontaneously, and many patients are asymptomatic. A short ECG recorded during a clinic appointment may therefore show sinus rhythm even if the same patient experiences AF at other times [1, 2]. This creates a motivation for longer-term monitoring and for low-burden screening systems that can collect rhythm evidence repeatedly outside the clinic.

ECG remains the diagnostic standard because it directly measures the electrical activity of the heart [1–3]. A sampled ECG waveform can be represented as

$$x[k] = x\left(\frac{k}{f_s}\right), \quad k = 0, 1, \dots, T - 1,$$

where f_s is the sampling frequency and T is the number of samples in the recording. In ECG, AF can be recognised using electrical rhythm markers such as irregular R–R intervals and absence of consistent discernible P-waves [1–3]. This is why ECG is the reference modality for clinical diagnosis and for rhythm labels in many benchmark datasets.

In this project, ECG is not used as an input to the final classifier. ECG is used indirectly through the dataset labels and rhythm annotations that define which PPG segments correspond to AF or sinus rhythm. The inference pipeline itself receives PPG-derived inputs: the filtered PPG waveform, the log-magnitude spectrogram computed from that waveform, and a 17-feature rhythm/quality vector extracted from PPG. This distinction is important. The project is therefore a PPG-based screening study using ECG-derived or clinically curated rhythm labels as reference information, not an ECG-and-PPG multimodal diagnostic model.

2.2 PPG as an Indirect Rhythm Signal

Photoplethysmography (PPG) is an optical technique for measuring peripheral blood volume changes [6]. A light source illuminates tissue and a photodetector measures changes in reflected or transmitted light caused by pulsatile blood flow. Each ventricular contraction produces a pressure wave that travels through the arterial system, which appears in PPG as a pulse waveform.

PPG is attractive for wearable screening because it is low cost, non-invasive, and already available in many smartwatches, fitness trackers, and clinical pulse sensors [4, 6, 11, 13]. Unlike wearable ECG, it does not usually require the user to actively complete an electrode circuit. This makes it suitable for passive repeated monitoring. Large wearable studies have shown that PPG-based irregular-rhythm notifications can identify previously unrecognised AF with high positive predictive value when alerting is conservative [11–13].

The same property that makes PPG convenient also makes it technically difficult. PPG does not measure atrial electrical activity. It cannot directly show P-waves, PR intervals, QRS complexes, or atrial fibrillatory activity. AF must instead be inferred from downstream mechanical evidence:

- irregular pulse-to-pulse intervals,
- variable pulse amplitudes caused by beat-to-beat stroke-volume changes,
- inconsistent pulse morphology,
- changes in spectral concentration and rhythm regularity.

These signals can be informative, but they are not specific to AF. Motion artefact, low perfusion, loose sensor contact, ambient light leakage, sensor pressure changes, vasoconstriction, missed pulse peaks, and duplicate detected peaks can all make a sinus-rhythm PPG segment appear irregular [4, 5, 14, 15]. Non-AF rhythms such as premature atrial contractions, premature ventricular contractions, atrial flutter with variable block, and bigeminy-like patterns can also produce irregular pulse intervals [15]. A robust PPG AF detector must therefore avoid the simple shortcut that “irregularity means AF”. It must judge whether the irregularity is physiologically plausible and whether the segment quality is high enough to support classification.

2.3 Digital Signal Representation and Preprocessing

The main dataset represents PPG as non-overlapping 30-second waveform segments sampled at 125 Hz. Each input segment therefore contains

$$T = 30 \times 125 = 3750, \quad x \in \mathbb{R}^{3750}.$$

The 30-second duration is clinically and technically motivated [1, 2, 4, 5]. Clinically, rhythm-monitoring studies commonly treat AF episodes lasting at least 30 seconds as meaningful. Technically, a 30-second window usually contains enough pulse intervals for rhythm variability features to become informative, while still being short enough to avoid mixing long periods of changing noise or rhythm state. Prior PPG AF work similarly indicates that approximately 20–30 seconds, or 25–40 beats, often provides enough rhythm evidence for discrimination [5].

Raw PPG segments contain baseline drift, high-frequency noise, and artefacts. The implemented preprocessing therefore applies band-pass filtering followed by z-score normalisation. The band-pass stage keeps the frequency range most relevant to plausible pulse waveforms while attenuating slow baseline drift and high-frequency noise:

$$0.5 \text{ Hz} \leq f \leq 8.0 \text{ Hz}.$$

After filtering, each segment is normalised as

$$x_{\text{norm}}[k] = \frac{x[k] - \text{mean}(x)}{\text{std}(x) + \varepsilon}.$$

This reduces dependence on absolute optical amplitude, which can vary with sensor placement, skin-device coupling, perfusion, and individual physiology. Normalisation is especially important for a neural waveform model because otherwise the model may learn patient- or device-specific amplitude shortcuts rather than rhythm and morphology evidence.

2.4 Pulse Peaks and Inter-Beat Interval Features

Many AF-relevant PPG features depend on reliable pulse peak detection. Let the detected pulse peak sample indices be

$$p_1, p_2, \dots, p_N,$$

where N is the number of detected peaks in a 30-second segment. The inter-beat interval (IBI) sequence in milliseconds is

$$\text{IBI}_i = 1000 \times \frac{p_i - p_{i-1}}{f_s}, \quad i = 2, \dots, N,$$

and the corresponding beat-to-beat heart rate is

$$\text{HR}_i = \frac{60000}{\text{IBI}_i}.$$

AF tends to increase the variability and unpredictability of the IBI sequence [1, 2, 4, 19]. The final feature vector therefore includes several standard rhythm-variability measures:

$$\text{SDNN} = \text{std}(\text{IBI}),$$

$$\text{RMSSD} = \sqrt{\text{mean} \left((\text{IBI}_i - \text{IBI}_{i-1})^2 \right)},$$

$$\text{pNN50} = \frac{\# \{ |\text{IBI}_i - \text{IBI}_{i-1}| > 50 \text{ ms} \}}{\# \{ \text{IBI differences} \}},$$

$$\text{CV}_{\text{IBI}} = \frac{\text{std}(\text{IBI})}{\text{mean}(\text{IBI})}.$$

These features encode complementary forms of irregularity. SDNN captures overall interval dispersion. RMSSD emphasises short-term beat-to-beat change. pNN50 measures the fraction of large successive interval differences. CV_{IBI} normalises interval variability by the mean interval, reducing dependence on absolute heart rate.

The feature vector also includes sample entropy, which measures the unpredictability of the interval sequence [4, 14]. In simplified form,

$$\text{SampEn}(m, r) = -\log \left(\frac{A}{B} \right),$$

where B counts matching subsequences of length m within tolerance r , and A counts matching subsequences of length $m + 1$. A more irregular sequence usually has fewer extended matches and therefore higher entropy. In PPG AF detection, entropy is useful because AF is not only variable but also less predictable than many regularly irregular patterns. However, entropy can also increase when peak detection fails, so it must be interpreted together with signal quality.

2.5 Morphology and Signal Quality Features

Pulse timing alone is not sufficient because artefact can create false irregularity [4, 5, 14, 15]. The implemented pipeline therefore also measures morphology and signal quality. Template correlation is used to quantify whether detected beats share a consistent pulse shape. Around each detected peak, a local beat window is extracted and compared with the average beat template. A high template correlation suggests repeated, physiologically plausible pulse morphology; a low value suggests motion artefact, poor contact, or unstable peak detection.

The pipeline also checks whether the detected pulse train is physiologically plausible. Very few peaks may indicate flat signal or missed detection, while too many peaks may indicate noise or duplicate detections. Heart-rate bounds reject implausible pulse rates. Short, long, or outlier-heavy IBI distributions are treated as unreliable because they often indicate peak-detection failure rather than true rhythm.

The final SQI gate combines several criteria:

- plausible peak count and heart-rate range,
- energy concentration in the expected heart-rate band,
- waveform distribution checks such as skewness and kurtosis,
- beat-template correlation,
- short, long, and outlier IBI fractions,
- zero-crossing behaviour,
- SNR-like quality,
- agreement between two peak-detection strategies.

This SQI stage creates an explicit no-call mechanism [4, 20]. Low-quality windows are rejected rather than forced into an AF/SR decision. This is important for PPG because a confident prediction on an unreliable segment can be more harmful than an abstention. The same quality information is also retained as model input through the 17-feature vector and is used later in quality-weighted record-level aggregation.

2.6 Frequency-Domain and STFT Features

PPG rhythm information is also visible in the frequency domain. A clean pulse waveform usually has energy concentrated near the heart-rate frequency and its harmonics. Motion artefact, poor contact, and broadband noise can spread energy across less physiologically meaningful frequency bands. This motivates both hand-crafted spectral features and a learned spectral branch in the neural model.

For a discrete signal $x[n]$, the discrete Fourier transform decomposes the signal into frequency components:

$$X[k] = \sum_{n=0}^{N-1} x[n] \exp\left(-j \frac{2\pi kn}{N}\right).$$

A full Fourier transform summarises which frequencies are present but does not show when those frequencies occur. The short-time Fourier transform (STFT) addresses this by applying the transform over overlapping windows:

$$X[m, k] = \sum_n x[n] w[n - mH] \exp\left(-j \frac{2\pi kn}{N_{\text{fft}}}\right),$$

where w is the analysis window, H is the hop length, m is the time-frame index, and k is the frequency-bin index. The implemented spectral branch uses

$$S[m, k] = \log(1 + |X[m, k]|).$$

This log-magnitude representation compresses large amplitude differences and gives the 2D convolutional spectral encoder a compact time-frequency view of the segment.

Spectral entropy is also included as a hand-crafted feature. If A_k denotes spectral energy in frequency bin k , a normalised spectral distribution can be written as

$$P_k = \frac{A_k}{\sum_j A_j}.$$

The spectral entropy is

$$H = - \sum_k P_k \log(P_k).$$

Low spectral entropy indicates energy concentrated in a small number of bands, while high spectral entropy indicates a broader or more disordered spectrum. In this project, spectral entropy is not used alone as an AF marker; it is used as one part of the rhythm/quality feature set and is complemented by the learned STFT branch.

2.7 Machine Learning and Deep Learning Approaches for PPG AF Detection

Existing PPG AF detection methods can be grouped into three broad families [4]. The first family is rule-based or feature-based detection. These methods detect pulse peaks, compute interval statistics such as SDNN, RMSSD, pNN50, entropy, and Poincaré-style descriptors, and then apply thresholds or simple decision rules. They are lightweight and interpretable, which makes them attractive for wearables, but they depend strongly on peak-detection quality [14].

The second family uses classical supervised machine learning over hand-crafted features. Models such as support vector machines, random forests, logistic models, and related tabular classifiers can combine multiple rhythm and quality features into a learned decision boundary. This is useful when no single feature is sufficient. For example, high interval variability may be more suspicious when the segment also has plausible peak count and good morphology consistency, but less trustworthy when template correlation and detector agreement are poor. The limitation is that tabular models can only use information captured by the engineered features.

The third family uses deep learning. One-dimensional CNNs can learn local pulse morphology from raw PPG waveforms. Sequence models such as recurrent networks or Transformers can compare rhythm evidence across time. Spectrogram-based CNNs can learn frequency-domain structure. Deep learning is useful because not every relevant PPG pattern is easily summarised by hand-crafted features. However, deep models can overfit to dataset-specific acquisition conditions, noise patterns, or patient context if signal quality and evaluation design are not handled carefully [21].

The final model in this project deliberately combines these ideas rather than choosing only one. The waveform branch learns morphology directly from the filtered PPG signal. The spectral branch learns time-frequency structure from the log-magnitude STFT. The feature branch receives explicit rhythm and signal-quality variables. This hybrid design is appropriate for PPG because AF evidence is distributed across pulse timing, pulse morphology, spectral structure, and segment reliability.

2.8 Hybrid Fusion and Record-Level Screening Logic

The implemented classifier, RhythmMorphologyFusionNet, estimates the probability of AF for each accepted 30-second segment:

$$p(\text{AF} \mid x, f) = \text{sigmoid}(z),$$

where x is the PPG waveform, f is the 17-dimensional feature vector, and z is the model logit. The model uses three encoders:

$$h_{\text{time}} = \phi_{\text{time}}(x), \quad h_{\text{spec}} = \phi_{\text{spec}}(x), \quad h_{\text{feat}} = \phi_{\text{feat}}(f).$$

These embeddings are concatenated and passed through a learned gate. In the implementation, the gate can also be conditioned on selected SQI-related features:

$$h = [h_{\text{time}}, h_{\text{spec}}, h_{\text{feat}}], \quad c = \text{selected SQI conditioning features},$$

$$g = \text{sigmoid}(W_2 \text{ GELU}(W_1[h, c])), \quad h_{\text{gated}} = h \odot g.$$

The gate allows the model to change the contribution of waveform, spectral, and feature evidence per segment. This is useful because the most reliable evidence source can vary. A clean segment may benefit strongly from waveform morphology and IBI features, while a noisier accepted segment may require more reliance on quality and spectral evidence.

The final screening decision is not made from a single isolated segment. Segment probabilities belonging to the same `record_id+event_id` group are aggregated using quality-weighted averaging:

$$P_{\text{record}} = \frac{\sum_i q_i p_i}{\sum_i q_i},$$

where p_i is the model probability for segment i and q_i is its quality score. This converts 30-second predictions into a record-event level decision and reduces the effect of isolated false-positive windows. This is closer to a practical wearable alerting system, where an alert should be supported by repeated evidence rather than one noisy window.

2.9 Design Requirements for This Project

The technical background leads to the following design requirements. First, the classifier should use PPG-derived inputs only at inference time, while treating ECG as the clinical reference behind the labels. Second, the system should operate on short windows that contain enough beats for rhythm analysis; this motivates 30-second segments. Third, peak detection and rhythm features should be combined with morphology and spectral evidence because interval irregularity alone is not specific to AF. Fourth, SQI should be explicit so that unreliable windows can be rejected or down-weighted. Fifth, evaluation should emphasise record-level screening behaviour, sensitivity, specificity, precision, F1, AUROC, and AUPRC rather than accuracy alone.

These requirements shape the rest of the report. Chapter 3 introduces the datasets used in the study. Chapter 4 describes the implemented preprocessing, peak detection, SQI, and feature extraction pipeline, and Chapter 5 presents the hybrid neural architecture and training objective. Chapter 6 explains the evaluation protocol, Chapter 7 reports both segment-level and record-level results, and Chapter 8 summarises the embedded deployment path. Chapters 9 to 12 then interpret the findings, state the main limitations, outline future work, and conclude the report.

2.10 Critical Positioning Against Prior PPG AF Work

Prior PPG-based AF work motivates this project but also shows why conservative claims are necessary. Large wearable studies demonstrate that passive pulse monitoring can identify AF-like irregularity at scale, but these systems are usually designed to issue notifications only when evidence is sustained and signal quality is acceptable. The Apple Heart Study reported high positive predictive value for irregular-pulse notifications when compared with subsequent ECG-patch evidence, but the notification logic was conservative and the study did not claim that PPG alone replaces diagnostic ECG [11]. The Fitbit Heart Study similarly used repeated irregular tachograms and ECG-patch

confirmation, reporting high PPV for concurrent AF among participants who had another irregular-rhythm detection during ECG monitoring [13]. These studies support the feasibility of wearable screening, but they also reinforce the need for confirmation, quality control, and aggregation over time.

The review by Pereira et al. describes the main algorithmic families used for PPG-based AF detection: pulse-interval features, classical machine learning, and deep learning [4]. The present project sits between these families. It does not rely only on hand-crafted rhythm features, because pulse timing can be corrupted by artefact. It also does not rely only on a black-box waveform network, because explicit rhythm and quality variables are useful for interpretability and embedded deployment. The hybrid design is therefore a response to a specific property of PPG: AF evidence is distributed across timing, morphology, spectrum, and reliability.

Compared with consumer wearable studies, this thesis uses ICU-derived data rather than free-living smartwatch data. This has both advantages and disadvantages. ICU data provide rich continuous physiological waveforms and clinically charted rhythm context, but ICU patients may have abnormal perfusion, comorbidities, medications, monitoring artefacts, and sensor conditions that differ from consumer wearable settings. Therefore, good performance on MIMIC-III-Ext-PPG p00–p02 is best interpreted as evidence of retrospective feasibility on a challenging clinical waveform benchmark, not as evidence of final deployment performance in free-living users.

2.11 Why Signal Quality Is a First-Class Model Component

Signal quality assessment is not a cosmetic preprocessing step in PPG AF detection. In PPG, a corrupted signal can mimic the same irregularity pattern that the model is trying to detect. This creates a failure mode in which an algorithm becomes good at recognising measurement failure rather than rhythm abnormality. SQI methods for ECG and PPG have therefore been proposed to decide whether reliable heart-rate or pulse information can be extracted from a window [20]. MIMIC-III-Ext-PPG also includes signal-quality assessments and describes correlation-based, beat-consistency, and template-matching approaches as part of its dataset construction [10].

The contribution of this project is not that it invents SQI from scratch, but that it integrates SQI into the whole screening logic. SQI is used before classification to reject unreliable windows, during modelling as part of the 17-feature vector and loss weighting, and after classification as part of quality-weighted aggregation. This makes the final decision a function not only of AF probability, but also of the reliability of the windows used to produce that probability.

2.12 Screening Versus Diagnosis in Clinical Guidance

Clinical guidance reinforces the distinction between an irregular pulse and a diagnosis of AF. NICE recommends manual pulse palpation when AF is suspected, but a 12-lead ECG is required to make the diagnosis if an irregular pulse is detected [1–3]. The same principle applies to PPG: the optical pulse signal can provide useful screening evidence, but it cannot directly show atrial electrical activity. This is why the wording throughout this thesis should remain conservative. The model detects AF-like pulse patterns in accepted PPG windows; it does not diagnose AF by itself.

This distinction is also important for threshold choice. A high-sensitivity operating point might be appropriate for clinician-facing retrospective review, where false positives can be triaged by ECG. A high-specificity operating point might be better for user-facing alerts, where repeated false positives could create anxiety and reduce trust. The final threshold in this project is therefore a validation-selected screening operating point, not a universal clinical threshold.

2.13 Evidence Map for the Thesis Contribution

Table 2.1 summarises how key prior work relates to the thesis. The purpose of this table is not to rank studies directly, because their populations, labels, sensors, and alert policies differ. Instead, it clarifies which gap this project addresses.

Source	Setting	Relevant contribution	Relation to this thesis
Pereira et al. [4]	Review of PPG AF methods	Summarises signal processing, ML, DL, and clinical challenges for PPG AF detection	Motivates hybrid timing/morphology/spectral design and conservative clinical framing
Perez et al. [11]	Large smartwatch study	Shows scalable irregular-pulse notification with ECG-patch follow-up	Supports record-level and conservative alert logic rather than single-window decisions
Lubitz et al. [13]	Large wearable PPG study	Uses repeated irregular tachograms and ECG-patch confirmation	Reinforces sustained-evidence and confirmatory-ECG framing
Orphanidou et al. [20]	Wearable ECG/PPG SQI	Uses SQI to assess whether reliable HR can be obtained	Supports treating SQI as part of decision reliability
MIMIC-III-Ext-PPG [9, 10]	ICU-derived benchmark	Large quality-assessed PPG benchmark with rhythm chart-event annotations	Provides main retrospective benchmark, but with ICU and chart-label validity caveats
This project	MIMIC PERform AF and MIMIC-III-Ext-PPG p00-p02	SQI-aware hybrid model, record-event aggregation, and embedded feature-model proof of concept	Provides an end-to-end MEng system study rather than only an isolated classifier

Table 2.1: Positioning of this thesis against selected PPG AF and signal-quality evidence.

Chapter 3

Datasets

3.1 MIMIC PERform AF: Controlled Baseline

MIMIC PERform AF contains 20-minute ECG and PPG recordings from critically ill adults. In this project, 35 records are used: 19 AF and 16 non-AF. Signals are segmented into 30-second windows, producing approximately 40 windows per 20-minute record. The dataset is used as a controlled baseline because it is small and binary. It is suitable for checking whether the preprocessing and classifier pipeline can separate AF from non-AF under relatively curated conditions, but it is not large enough to support strong generalisation claims.

The dataset is part of the MIMIC PERform datasets, which were extracted from the MIMIC-III Waveform Database and released for PPG beat-detection benchmarking and related physiological signal analysis [7, 8].

Table 3.1: Patient-level split for the MIMIC PERform AF baseline experiment.

Split	Records	AF records	Non-AF records
Train	25	13	12
Validation	5	3	2
Test	5	3	2

3.2 MIMIC-III-Ext-PPG v1.1.0: Main Dataset

The main experiment uses MIMIC-III-Ext-PPG v1.1.0, published on PhysioNet on 17 March 2026 [9]. The resource is a large-scale, annotated, and quality-assessed PPG benchmark derived from the MIMIC-III Waveform Database Matched Subset and MIMIC-III clinical records. The associated publication presents the broader dataset design and annotation process [10].

Table 3.2: Key properties of the full heart-rhythm task in MIMIC-III-Ext-PPG v1.1.0.

Property	Value
Patients	6,189
Non-overlapping 30-second PPG segments	6,399,754
Total duration	approximately 53,331 hours
Sampling rate	125 Hz
Mean age	64.1 ± 17.0 years
Mean weight	82.2 ± 22.6 kg
Mean height	169.5 ± 10.5 cm
Female patients	43.9%

The dataset contains harmonised rhythm labels including SR, sinus tachycardia, AF, sinus bradycardia, ventricular pacing, first-degree AV block, atrioventricular pacing, atrial flutter, atrial pacing,

bundle branch block, and less common rhythms. The full label distribution includes 3,950,724 SR segments and 597,769 AF segments. Each segment is stored in WFDB format. PPG is present for all segments, and simultaneous ECG lead II, arterial blood pressure, and respiratory signals are included where available. The metadata also include demographics, clinical information system, ICD-derived diagnosis codes, 10-fold stratification labels, derived HR/RR/BP annotations, and SQI values. The released waveform signals are raw segmented signals; users are responsible for applying filtering, denoising, and normalisation before model training.

3.3 Scope Used in This Project

This project uses prefixes p00, p01, and p02 of MIMIC-III-Ext-PPG. This subset was chosen because it is already much larger than MIMIC PERform AF and was computationally feasible within the project time and available GPU resources. The binary task retains AF segments as the positive class and SR segments as the negative class. Other rhythm classes are not included in the final binary results. This makes the task interpretable as AF versus sinus rhythm rather than general multi-class arrhythmia classification.

Table 3.3: Fold-based split for the p00–p02 AF-versus-SR subset used in this project.

Split	Metadata folds	Accepted segments	SR segments	AF segments	AF fraction	Record-event groups	SR groups	AF groups	Subjects
Train	0–7	622,065	537,476	84,589	0.1360	41,376	33,943	7,433	757
Validation	8	93,200	82,745	10,455	0.1122	5,742	4,848	894	97
Test	9	74,599	67,847	6,752	0.0905	5,301	4,613	688	103
Total	0–9	789,864	688,068	101,796	0.1289	52,419	43,404	9,015	957

Record-level evaluation uses the combined group key `record_id+event_id` because a single waveform record may contain more than one rhythm chart event. This avoids assigning conflicting labels to the same evaluation group.

Chapter 4

Signal Processing and Feature Extraction

4.1 Segmentation

All signals are represented as non-overlapping 30-second windows [1, 2, 4, 5, 9, 10]. At 125 Hz, each PPG segment contains 3,750 samples. The 30-second duration is clinically meaningful because AF episodes of at least 30 seconds are commonly considered significant, and it provides enough beats for IBI-derived rhythm features to be informative [1, 2, 4, 5].

For MIMIC-III-Ext-PPG, the source dataset already provides 30-second WFDB segments. The project still applies its own filtering, normalisation, SQI processing, and feature extraction because the released waveforms are raw segmented signals.

4.2 Filtering and Normalisation

Each PPG segment is filtered using a fourth-order band-pass filter. The retained band of 0.5–8.0 Hz keeps the frequency range relevant to plausible pulse rates and suppresses slow baseline drift and high-frequency noise. After filtering, each segment is z-score normalised so that the neural model receives amplitude-standardised waveform inputs. This ordering follows standard PPG signal-processing practice, in which preprocessing is performed before peak detection, SQI assessment, and downstream feature extraction [22].

Table 4.1: Filtering and segmentation parameters.

Parameter	Value
Low cut-off	0.5 Hz
High cut-off	8.0 Hz
Filter order	4
Segment length	30 s
Segment stride	30 s
Normalisation	z-score

4.3 Peak Detection

Pulse peaks are detected using an Elgendi-inspired morphology and derivative pipeline [22, 23]. The detector combines smoothed PPG, first- and second-derivative cues, local maxima, zero-crossing structure, amplitude constraints, prominence thresholds, and a minimum peak distance derived from plausible heart-rate limits. An alternate prominence-based detector is also used as a consistency check. Low agreement between detectors suggests that the detected pulse train may not be reliable.

Table 4.2: Peak-detector parameters.

Parameter	Value
Minimum HR	35 bpm
Maximum HR	220 bpm
Prominence scale	0.3
Minimum absolute prominence	0.02
Peak refinement radius	0.2 s

4.4 SQI Acceptance Criteria

The SQI gate rejects segments that are unlikely to support reliable rhythm analysis. The criteria include peak-count limits, spectral concentration in the heart-rate band, waveform distribution checks, template similarity, IBI plausibility, zero-crossing rate, SNR-like quality, and detector agreement.

Table 4.3: SQI criteria used to accept or reject segments.

SQI criterion	Threshold used
Minimum peak count	15
Maximum peak count	125
Minimum HR	35 bpm
Maximum HR	220 bpm
Minimum heart-band energy ratio	0.55
Maximum absolute skewness	2.0
Maximum kurtosis	12.0
Minimum template correlation	0.55
Maximum short IBI fraction	0.20
Maximum long IBI fraction	0.20
Maximum IBI outlier fraction	0.35
Maximum zero-crossing rate	0.20
Minimum SNR SQI	1.2
Minimum detector agreement	0.55

Accepted segments are stored in `ppg_accepted_segments.npz` and summarised in `ppg_accepted_segment_summary.csv`. Rejected segments remain available in the full segment summary for auditing.

4.5 Hand-Crafted Feature Vector

The offline neural model and the compact Edge AI model use a 17-dimensional feature vector.

Missing feature values are filled using medians computed on the training split only. Features are then standardised using training-split means and standard deviations, preventing validation or test information from leaking into preprocessing.

Table 4.4: Hand-crafted rhythm and quality features used by the offline and embedded models.

Feature	Description
peak_count	Number of detected pulse peaks in the 30-second segment
heart_band_energy_ratio	Spectral energy concentration in the expected heart-rate band
signal_skewness	Distribution skewness of the filtered PPG segment
template_correlation	Similarity between individual beats and the average beat template
estimated_hr_bpm	Estimated heart rate from detected peaks
quality_score	Composite segment quality score
ibi_count	Number of inter-beat intervals
mean_ibi_ms	Mean IBI duration in milliseconds
median_ibi_ms	Median IBI duration in milliseconds
sdnn_ms	Standard deviation of IBIs
rmssd_ms	Root mean square of successive IBI differences
pnn50	Fraction of successive IBI differences greater than 50 ms
mean_hr_bpm	Mean beat-to-beat heart rate
std_hr_bpm	Standard deviation of beat-to-beat heart rate
cv_ibi	Coefficient of variation of IBIs
sample_entropy	Entropy measure of rhythm irregularity
signal_spectral_entropy	Entropy of the PPG spectral distribution

Chapter 5

Deep Learning Model

5.1 Problem Formulation

Each input sample consists of a filtered and normalised 30-second PPG waveform x with 3,750 samples, a 17-dimensional feature vector f , and a binary label y , where $y = 1$ denotes AF and $y = 0$ denotes SR. The model outputs a logit z , and the predicted AF probability is

$$p(\text{AF} \mid x, f) = \sigma(z).$$

The classification threshold is selected on the validation set and then fixed for the test set.

5.2 Model Selection Rationale and Mathematical Background

The model was chosen to match the structure of the PPG AF detection problem rather than to maximise architectural complexity. A 30-second PPG segment contains several complementary signals: local pulse morphology, beat-to-beat timing irregularity, frequency-domain concentration or noise, and explicit signal-quality indicators. A feature-only model can capture rhythm irregularity but may miss waveform shape. A pure waveform CNN can learn morphology but may underuse clinically interpretable pulse-rate variability features. A Transformer-only model is also more expensive than necessary for local pulse-shape extraction. The retained hybrid model therefore uses separate branches for morphology, spectrum, and hand-crafted rhythm/quality features before fusing them.

Formally, each segment is represented as

$$x \in \mathbb{R}^T, \quad T = 3750, \quad f \in \mathbb{R}^{17}, \quad y \in \{0, 1\}.$$

The classifier estimates the posterior probability of AF using

$$z = \psi(\phi_{\text{time}}(x), \phi_{\text{spec}}(x), \phi_{\text{feat}}(f)),$$

where ϕ_{time} is the time-domain waveform encoder, ϕ_{spec} is the spectral encoder, ϕ_{feat} is the feature encoder, and ψ is the gated fusion classifier.

Several AF-relevant features are already well described mathematically by rhythm statistics:

$$\begin{aligned} \text{SDNN} &= \text{std}(\text{IBI}), & \text{RMSSD} &= \sqrt{\text{mean}((\text{IBI}_i - \text{IBI}_{i-1})^2)}, \\ \text{pNN50} &= \frac{\#\{|\text{IBI}_i - \text{IBI}_{i-1}| > 50 \text{ ms}\}}{\#\{\text{IBI differences}\}}, & \text{CV}_{\text{IBI}} &= \frac{\text{std}(\text{IBI})}{\text{mean}(\text{IBI})}. \end{aligned}$$

These features directly encode irregularity, while quality features such as template correlation and heart-band energy help the classifier decide whether the irregularity is likely physiologic or artefactual.

5.3 RhythmMorphologyFusionNet Overview

The offline classifier is RhythmMorphologyFusionNet. It combines three complementary evidence streams: time-domain pulse morphology from the raw PPG waveform, frequency-domain rhythm and noise structure from a log-magnitude spectrogram, and explicit pulse-rate variability and quality features from the 17-dimensional vector. Each branch produces a 64-dimensional embedding. These embeddings are concatenated, passed through a learned gate, and classified by a multilayer head.

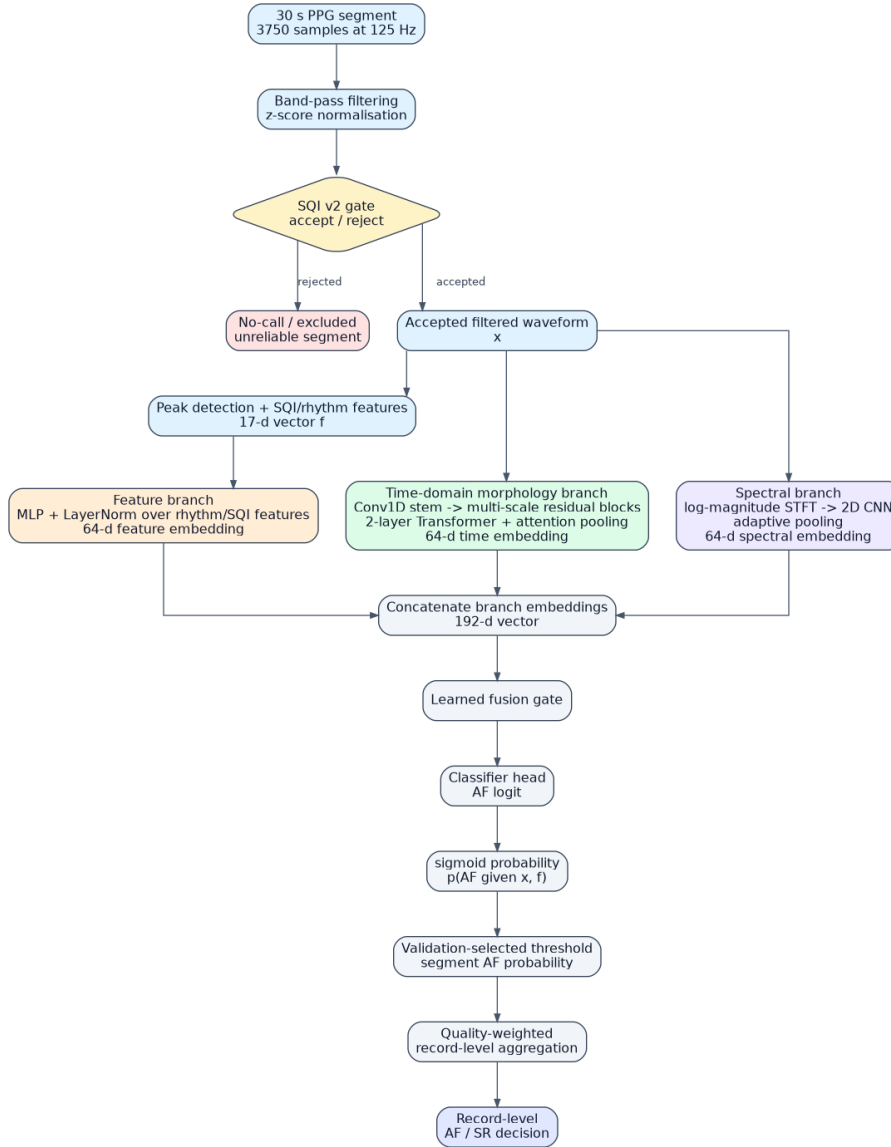


Figure 5.1: RhythmMorphologyFusionNet combines SQI gating, a time-domain morphology branch, a spectral branch, and a 17-feature rhythm/SQI branch. Accepted segment probabilities are converted into record-level AF/SR decisions using quality-weighted aggregation.

5.4 Time-Domain Branch

The time-domain branch receives the waveform as a single-channel sequence. It begins with a convolutional stem followed by three multi-scale residual blocks and a compact two-layer Transformer encoder that summarises approximately 235 temporal tokens [24].

Table 5.1: Time-domain convolutional stack.

Block	Input channels	Output channels	Stride
Convolutional stem	1	32	2
Multi-scale block 1	32	64	2
Multi-scale block 2	64	96	2
Multi-scale block 3	96	128	2

Each multi-scale residual block contains three parallel convolutional branches.

Table 5.2: Parallel convolutions inside each multi-scale residual block.

Branch	Kernel size	Dilation	Purpose
1	3	1	Local pulse shape
2	7	2	Medium-range morphology
3	15	3	Longer pulse and rhythm context

After the convolutional stack, the tokens are processed by a compact Transformer encoder.

Table 5.3: Transformer encoder parameters for the time-domain branch.

Transformer parameter	Value
Model dimension	128
Attention heads	4
Feed-forward dimension	512
Layers	2
Dropout	0.1
Activation	GELU
Normalisation style	pre-norm

Attention pooling summarises the token sequence:

$$a_t = w^\top \tanh(Wh_t), \quad \alpha_t = \frac{\exp(a_t)}{\sum_j \exp(a_j)}, \quad h_{\text{time}} = \sum_t \alpha_t h_t.$$

The pooled 128-dimensional representation is finally projected to a 64-dimensional embedding.

5.5 Spectral Branch

The spectral branch computes a log-magnitude short-time Fourier transform from the same waveform. This branch helps the model capture periodicity, dominant pulse-rate bands, broadband noise, and spectral irregularity.

The resulting spectrogram is passed through a compact 2D CNN with three convolutional layers, group normalisation, GELU activations, max pooling, global average pooling, and a final linear projection to a 64-dimensional spectral embedding.

5.6 Feature Branch

The feature branch embeds the 17 hand-crafted rhythm and quality features using a two-layer multilayer perceptron consisting of a 17-to-64 projection, layer normalisation, GELU activation,

Table 5.4: STFT parameters for the spectral branch.

STFT parameter	Value
FFT size	128
Window length	128
Hop length	32
Window	Hann
Transform	$\log_{10}(\text{abs}(\text{STFT}))$

dropout, and a second 64-to-64 transformation. This branch gives the model direct access to clinically interpretable rhythm variability information such as RMSSD, pNN50, and sample entropy, as well as signal-quality cues such as template correlation and heart-band energy.

5.7 Gated Fusion and Classifier Head

The three 64-dimensional embeddings are concatenated to form

$$h = [h_{\text{time}}; h_{\text{spectral}}; h_{\text{feature}}] \in \mathbb{R}^{192}.$$

A learned gate is then computed from the concatenated representation:

$$g = \sigma(W_2 \text{GELU}(W_1 h)).$$

The gate is split into three 64-dimensional sub-gates and applied separately to the branch embeddings. This mechanism allows the model to change the relative contribution of waveform morphology, spectral evidence, and hand-crafted rhythm features on a per-sample basis. The final classifier head maps the gated 192-dimensional representation through two hidden layers before producing a single AF logit.

5.8 Training Objective

The final large-scale model uses a quality-aware focal loss that combines binary cross-entropy with positive-class weighting, focal modulation, label smoothing, and signal-quality weighting [25]. Label smoothing is applied as

$$\tilde{y} = y(1 - 0.02) + 0.5 \times 0.02.$$

Focal modulation uses $\gamma = 1.5$, and the segment quality score q enters as a sample weight,

$$w_q = 0.6 + 0.4q.$$

This means that all accepted segments can contribute to training, but higher-quality segments contribute slightly more strongly.

5.9 Class Imbalance Handling

The final SQI run uses a weighted random sampler. The raw training split contains 84,589 AF segments and 537,476 SR segments after SQI acceptance. Instead of training on this raw class ratio, the sampler targets an AF fraction of 0.333333, corresponding approximately to an AF:SR ratio of 1:2 in sampled mini-batches. This choice preserves more SR diversity than strict 1:1 balancing while still increasing AF exposure during optimisation.

Table 5.5: Final training configuration for the final SQI p00-p02 run.

Parameter	Value
Device	CUDA
Epoch limit	30
Early stopping patience	10
Epochs run	18
Best epoch	8
Batch size	32
Learning rate	3e-4
Weight decay	5e-5
Optimiser	AdamW [26, 27]
Scheduler	CosineAnnealingLR
Loss	Quality-aware focal loss
Focal gamma	1.5
Label smoothing	0.02
Balanced sampler	yes
Sampler AF fraction	0.333333
Mixup	disabled
Automatic mixed precision	disabled
Threshold objective	validation record-level F1

5.10 Data Augmentation and Test-Time Augmentation

During training, waveform augmentation is available for the training split. The implemented augmentations include amplitude scaling, Gaussian noise, circular time shift, low-frequency drift, local masking, and mild time warping. These operations are intended to reduce overfitting to exact waveform shape and improve robustness to realistic PPG variability.

During evaluation, test-time augmentation averages predictions over three temporal shifts,

$$\text{shifts} = (0, -8, 8).$$

This makes the prediction less sensitive to very small alignment changes in the 30-second window.

5.11 Record-Level Aggregation

The model produces segment-level probabilities, but the primary evaluation is record-level. Segment probabilities belonging to the same `record_id+event_id` group are aggregated using a quality-weighted average,

$$p_{\text{record}} = \frac{\sum_i q_i p_i}{\sum_i q_i},$$

with the arithmetic mean used if all quality weights are zero. This aggregation is important because an AF screening alert should be based on an episode or recording rather than a single isolated 30-second prediction.

5.12 Threshold Selection

The final decision threshold is selected using validation record-level F1. For the final SQI model, the selected threshold is 0.713. This threshold is fixed before test evaluation. Test-set threshold sweeps are reported only as analysis and are not used as the primary result.

5.13 Dataset Provenance and Label Validity

The dataset choice is a major strength of this project, but it also determines the interpretation of the results. MIMIC-III-Ext-PPG is derived from the MIMIC-III Waveform Database Matched Subset and associated clinical records. The full resource provides millions of 30-second PPG segments, multiple rhythm labels, and signal-quality information [9, 10]. This makes it much more suitable for large-scale retrospective model development than small binary datasets. However, the heart-rhythm labels are derived from clinical chart events rather than from prospective smartwatch-style adjudication for every 30-second window. The label should therefore be understood as a strong reference annotation for retrospective benchmarking, but not as the same thing as a cardiologist-adjudicated ECG label assigned independently to every segment.

This matters because the project uses PPG alone at inference time. The model is not allowed to use the simultaneously available ECG, ABP, or respiratory signals as inputs, even though some of those modalities are available in the dataset. The ECG and charted rhythm information serve as reference context behind the labels. This design keeps the task aligned with the intended application: screening from PPG when ECG is not continuously available. At the same time, it creates an unavoidable validity constraint. If a charted rhythm event is imprecisely timed relative to a PPG window, the training label may be noisy. The record-event aggregation strategy partly reduces sensitivity to individual mislabeled or transitional windows, but it cannot eliminate all label-timing uncertainty.

The use of MIMIC PERform AF should also be interpreted carefully. It is useful as a controlled sanity check because it has 35 critically ill adults labelled as AF or non-AF and contains ECG, PPG, and respiration signals [7, 8]. A perfect record-level result on five held-out records confirms that the pipeline can separate clear binary cases in a small controlled setting. It does not provide the main evidence for generalisation. The larger p00–p02 MIMIC-III-Ext-PPG experiment is therefore the main result, while MIMIC PERform AF is retained only as a baseline validation stage.

5.14 Defined Scope of the p00–p02 Experiment

The p00–p02 scope should be presented as a defined experimental boundary rather than as a hidden shortcut. After SQI acceptance, this subset still contains hundreds of thousands of 30-second windows and tens of thousands of record-event groups, which is large for an MEng individual project. It is therefore sufficient to test the engineering pipeline, model training, aggregation strategy, and deployment path under substantial data volume. However, it is not equivalent to validating across the entire MIMIC-III-Ext-PPG release. Broader evaluation could reveal prefix-specific variation caused by differences in patients, recording conditions, rhythm composition, or data ordering. The thesis should therefore claim retrospective feasibility on p00–p02 and list broader evaluation as the next validation step.

The binary AF-versus-SR scope is similarly defensible but limited. A binary task isolates the core question of whether AF-like irregularity can be distinguished from sinus rhythm in accepted PPG windows. This is a reasonable first engineering target because it allows clear interpretation of sensitivity, specificity, precision, and F1. However, excluding other rhythm classes removes important confounders. A future screening system must distinguish AF from atrial flutter, ectopy, pacing, bradycardia, tachycardia, heart block, and other patterns that can distort pulse timing. The current binary result should therefore be treated as an AF-versus-SR screening result, not as evidence that the model can manage the full clinical arrhythmia spectrum.

5.15 Data Leakage and Evaluation Discipline

Retrospective physiological-signal studies are vulnerable to several forms of leakage. The most obvious leakage would occur if normalisation parameters, imputation medians, threshold choices, or calibration decisions were computed using validation or test labels. This project avoids that by computing feature-imputation medians and standardisation statistics on the training split only, selecting the operating threshold on validation record-level F1, and then fixing that threshold before test evaluation.

A second risk is grouping leakage. Multiple 30-second windows from the same record-event are highly correlated, so treating them as independent clinical alerts would overstate the amount of evidence. The project addresses this by using segment-level metrics only as diagnostic analysis and by making record-event aggregation the primary result. The key record identifier is the combined `record_id+event_id`, which avoids assigning one mixed rhythm event to conflicting evaluation units.

A third risk is within-subject dependence. A held-out test set can contain many record-events but fewer unique subjects. This is why the subject-clustered bootstrap intervals are important. They are wider than ordinary record-event bootstrap intervals, showing that repeated record-events from a smaller number of people can make uncertainty appear smaller than it really is. This does not invalidate the test result, but it gives a more cautious estimate of uncertainty and supports the need for external validation.

5.16 Representativeness and Domain Shift

The ICU context is both a benefit and a limitation. It provides high-volume monitored waveforms and clinically relevant rhythm events, but it differs from the intended wearable or near-sensor deployment environment. ICU PPG may be affected by vasopressors, oedema, poor peripheral perfusion, arterial lines, movement during care, sensor replacement, and acute illness. Consumer wrist PPG may instead be affected by exercise, device looseness, skin-sensor contact, ambient light, temperature, and daily motion. A model that performs well in one domain may not transfer unchanged to the other.

This is why the deployment claim is phrased as a feasibility path rather than deployed clinical performance. The retrospective model demonstrates that the algorithmic pipeline can learn from accepted PPG windows in a large clinical benchmark. A future hardware study must then test whether the same features and thresholds remain stable on the target sensor, with synchronised ECG confirmation and explicit measurement of no-call coverage.

5.17 Rationale for the Three-Branch Architecture

The three-branch architecture is motivated by the fact that no single representation of PPG is sufficient for robust AF screening. The feature branch captures explicit rhythm variability and quality measures, which are interpretable and compact. The time-domain branch captures waveform morphology, including pulse shape, local deformation, and repeated beat structure that may not be captured by scalar features. The spectral branch captures frequency concentration, broadband noise, and rhythm-related time-frequency patterns. The gated fusion mechanism gives the model a way to combine these sources differently for different windows.

This rationale should not be overstated. The neural branch ablation indicates that the waveform-only model was very competitive in one retrained run, and the independently retrained full-fusion model did not dominate all branch variants. The correct interpretation is therefore not that gated fusion is proven universally superior. Rather, the architecture is a plausible system design that allows complementary evidence to be used, while the current ablation suggests that learned time-domain morphology is particularly important. This balanced interpretation is stronger than claiming an architectural novelty that the ablation does not fully prove.

5.18 SQI as Abstention Rather Than Filtering Only

In a deployment setting, SQI should be interpreted as an abstention mechanism. A low-quality window is not necessarily SR and not necessarily AF; it is an unreliable measurement. This distinction is important because forcing every window into a binary label can produce false confidence. The final pipeline therefore treats rejected windows as no-call evidence rather than negative evidence. This is closer to a safe monitoring workflow: when the signal is unreliable, the system should communicate poor signal quality or wait for more evidence.

The same principle applies to quality-weighted aggregation. A record-event decision should be influenced more by windows with plausible pulse morphology, reliable peak detection, and concentrated

heart-band energy than by windows near the acceptance boundary. The weighted average

$$P_{record} = \frac{\sum_i q_i p_i}{\sum_i q_i}$$

implements this logic directly. The aggregation ablation shows that quality-weighted averaging gives only a small improvement over an unweighted mean, but it is still preferable as a design choice because it encodes the intended reliability principle and is less vulnerable than max-style pooling to isolated high-probability windows.

5.19 Why Focal Loss and Class-Aware Sampling Are Used

The p00–p02 task is imbalanced, with AF representing a minority of accepted segments [25]. If the model were trained only on the raw class distribution, mini-batches would often contain many more SR than AF examples, and optimisation could favour specificity while under-learning minority-class decision boundaries. The weighted sampler partially corrects this by increasing AF exposure while retaining more SR diversity than strict one-to-one balancing. This matters because SR is not a single waveform pattern; it includes a range of heart rates, morphologies, and noise conditions.

The focal component of the loss gives additional emphasis to examples that remain difficult after the current model prediction. In this setting, hard examples may include AF windows with regular-looking pulse intervals, SR windows with ectopic-like irregularity, or accepted windows near the SQI boundary. The quality-aware weight then slightly increases the influence of higher-quality accepted windows. This does not mean low-quality accepted windows are ignored. Instead, the loss reflects the assumption that clear, physiologically plausible windows provide more reliable supervision than marginal accepted windows.

5.20 Methodological Alternatives Considered

Several simpler alternatives are natural baselines. A rule-based detector could threshold RMSSD, pNN50, sample entropy, or coefficient of variation. This would be compact and interpretable, but it would be brittle because artefact and ectopy can create similar interval irregularity. A feature-only logistic or tree-based model can combine multiple variables, and the logistic baseline in this project is strong. However, feature-only models depend entirely on the feature extractor and may miss waveform-shape evidence. A pure deep waveform model can learn morphology directly, but it may underuse known rhythm statistics and may be harder to translate into a small embedded model. Record-level multiple-instance learning is conceptually attractive, but the exploratory MIL variants in this project did not improve the selected operating point.

The retained system is therefore a pragmatic compromise. It uses a strong explicit feature representation for interpretability and deployment, a waveform branch for learned morphology, a spectral branch for rhythm/noise structure, and record-level aggregation for screening behaviour. The final claim should be that this combination produced the best retained operating point among the completed experiments, not that it is the only possible or theoretically optimal architecture.

5.21 Offline–Embedded Model Separation

The offline and embedded models answer different questions. The offline RhythmMorphologyFusionNet asks how well a hybrid PPG model can perform retrospectively when waveform, STFT, and feature evidence are available. The embedded Nordic Edge AI model asks whether the compact 17-feature representation can be passed through a generated runtime and executed in Zephyr firmware. These are related but not equivalent. Table 5.6 makes this distinction explicit.

Aspect	Offline hybrid model	Embedded proof-of-concept model
Purpose	Retrospective performance evaluation	Deployment-path feasibility
Inputs	Filtered waveform, log-magnitude STFT, 17 features	17 precomputed float32 features
Model type	Multi-branch neural fusion model	Compact Nordic Edge AI / Neuton CPU model
Main output	Segment probability aggregated to record-event probability	Firmware inference class output on holdout feature rows
Evidence provided	Screening performance on p00–p02 held-out test fold	Runtime integration and smoke-test feasibility
Evidence not provided	Prospective clinical validation	Full hybrid on-device performance or live feature extraction

Table 5.6: Separation between the retrospective research model and the embedded proof-of-concept model.

Chapter 6

Evaluation Protocol

6.1 Primary and Secondary Evaluation Levels

The primary evaluation is record-level because this better reflects an AF screening workflow. Segment-level evaluation is also reported because it is useful for diagnosing model behaviour on individual 30-second windows.

Table 6.1: Primary and secondary evaluation levels.

Level	Definition	Role in paper
Segment-level	Each 30-second window is classified independently	Secondary diagnostic analysis
Record-level	Segment probabilities are aggregated per <code>record_id+event_id</code>	Primary performance result

6.2 Metrics

The reported metrics are accuracy, sensitivity (recall), specificity, precision, F1, AUROC, and AUPRC. Because the task is imbalanced, accuracy is interpreted cautiously. The headline model-selection metric is record-level F1, because it balances AF detection sensitivity against the false-alert burden captured by precision. Sensitivity, specificity, precision, AUROC, and AUPRC are reported alongside F1 to show both the operating-point trade-off and threshold-independent ranking performance.

6.3 Threshold Selection and Operating-Point Interpretation

The model outputs probabilities rather than hard labels. A threshold is therefore needed to convert predicted AF probability into an AF/SR decision. In this project, the primary threshold is selected on the validation set by maximising record-level F1. This choice is appropriate because the final screening output is record-event level rather than isolated segment level, and because F1 balances sensitivity against precision in the presence of class imbalance.

The selected threshold is then fixed before test evaluation. Test-set threshold sweeps are included only for analysis. This distinction is important: a test-best threshold can show how the sensitivity-precision trade-off behaves, but it should not be reported as the primary model result because it uses information from the held-out test set. The main result therefore reflects the threshold that would have been chosen before seeing test labels.

6.4 Segment-Level, Record-Level, and Episode-Level Reasoning

The final implementation uses record-event groups as the practical episode-level unit because MIMIC-III-Ext-PPG provides 30-second segments associated with rhythm chart events and record identifiers. Aggregating by `record_id+event_id` avoids treating every 30-second window as an independent clinical alert. Segment-level metrics remain useful because they reveal local classifier behaviour. Record-level metrics are more important for the final claim because they show whether repeated evidence across a record-event supports a reliable screening decision. This mirrors wearable alert logic, where a system usually should not notify a user based on one noisy or isolated window.

6.5 False Positives, False Negatives, and Screening Risk

False positives and false negatives have different consequences in AF screening. False positives may create anxiety, unnecessary follow-up, or increased clinician workload. False negatives may delay detection of AF. Because PPG is a screening signal, neither error type can be ignored. This is why the paper reports precision, specificity, and sensitivity together. Precision is particularly important under class imbalance because even a small false-positive rate can produce many false alerts when the negative class is large.

6.6 Coverage and No-Call Behaviour

The SQI gate creates a form of coverage control. Rejected segments are not passed to the classifier as reliable evidence. In deployment terms, this corresponds to a no-call or unreliable-signal state. Such abstention can improve trust by avoiding forced predictions on corrupted PPG, but excessive abstention would reduce usefulness because the system would be silent too often. The current final results are reported on SQI-accepted segments and record-event groups derived from those segments. A future deployment study should therefore report both classification performance on accepted windows and coverage over all available time.

6.7 Reproducibility Artifacts

The final SQI run saved artifacts to the experiment output directory. Key output files include the following.

```
metrics.json
best_model.pt
training_history.csv
val_segment_predictions.csv
test_segment_predictions.csv
val_record_predictions.csv
test_record_predictions.csv
val_segment_threshold_sweep.csv
test_segment_threshold_sweep.csv
val_record_threshold_sweep.csv
test_record_threshold_sweep.csv
```

Chapter 7

Results

7.1 Controlled Baseline: MIMIC PERform AF

The MIMIC PERform AF baseline achieved perfect record-level results on its five-record test split. Segment-level results were also high, but not perfect because individual 30-second windows are noisier than record-level averages.

Table 7.1: MIMIC PERform AF test results.

Evaluation level	Accuracy	Sensitivity	Specificity	Precision	F1	AUROC	AUPRC
Segment-level test	0.9447	1.0000	0.8625	0.9154	0.9558	0.9980	0.9986
Record-level test	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 7.2: MIMIC PERform AF record-level confusion matrix.

	Predicted SR/non-AF	Predicted AF
True SR/non-AF	2	0
True AF	0	3

Table 7.3: MIMIC PERform AF segment-level confusion matrix at threshold 0.170.

	Predicted SR/non-AF	Predicted AF
True SR/non-AF	69	11
True AF	0	119

These results confirm that the signal processing and hybrid model are capable of identifying AF when the data are relatively controlled and binary. However, the dataset is too small for the result to be considered the main evidence.

7.2 Large-Scale Experiment Comparison

The main model-development comparison is performed on MIMIC-III-Ext-PPG p00–p02. Record-level metrics are used for the headline comparison.

Among the completed p00–p02 runs, the final SQI pipeline gives the strongest record-level F1 and precision. Compared with the feature-only logistic baseline, the final hybrid model improves record-level F1 from 0.7775 to 0.8217 and precision from 0.7214 to 0.8194, although the feature-only baseline has slightly higher sensitivity and AUROC. This suggests that the engineered 17-feature representation is already strong, while the retained hybrid checkpoint improves the selected screening operating point. Compared with the initial large-scale hybrid experiment, the final SQI pipeline improves accuracy, specificity, precision, F1, and AUPRC, while sensitivity decreases from 0.8454

Table 7.4: Headline experiment comparison at record level.

Experiment	Dataset / scope	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1	AUROC	AUPRC
Controlled baseline	MIMIC PERform AF	0.170	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Feature-only logistic baseline	MIMIC-III-Ext-PPG p00-p02	0.763	0.9374	0.8430	0.9514	0.7214	0.7775	0.9542	0.8142
Initial large-scale hybrid	MIMIC-III-Ext-PPG p00-p02	0.516	0.9435	0.8454	0.9582	0.7519	0.7959	0.9229	0.8165
Class-aware 1:2 sampler	MIMIC-III-Ext-PPG p00-p02	0.781	0.9514	0.8410	0.9680	0.7973	0.8186	0.9467	0.8311
Final SQI pipeline	MIMIC-III-Ext-PPG p00-p02	0.713	0.9536	0.8241	0.9729	0.8194	0.8217	0.9272	0.8365

to 0.8241. Because this comparison changes several ingredients at once, it should be interpreted as model-development evidence rather than as an isolated SQI ablation.

Additional exploratory record-level architectures were tested but not retained.

Table 7.5: Exploratory record-level models that were not retained as the final system.

Exploratory model	Record-level F1	AUROC	AUPRC	Interpretation
Hierarchical record model	0.7849	0.9486	0.7459	Did not improve over the SQI-aware hybrid
Record MIL	0.7558	0.9471	0.7611	Lower F1 and precision
Record MIL second variant	0.7864	0.9545	0.7799	Better AUROC, lower F1 than the final SQI model

7.3 Feature-Level SQI Ablation

To separate part of the SQI contribution from the full hybrid model, a feature-level ablation was run using weighted logistic regression. This ablation does not replace full retraining of the neural waveform/STFT/fusion model, but it tests SQI filtering, SQI-derived features, and quality-weighted aggregation under a consistent lightweight classifier. For each condition, the decision threshold was selected on validation record-level F1 and then fixed on the test set.

Table 7.6: Feature-level SQI ablation at record level on the p00-p02 test fold.

Feature-level condition	SQI filtering	SQI features	Quality-weighted aggregation	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1	AUROC	AUPRC
Rhythm features only, no SQI filter	No	No	No	0.845	0.9507	0.8266	0.9693	0.8011	0.8137	0.9521	0.8126
All 17 features, no SQI filter	No	Yes	No	0.783	0.9413	0.8324	0.9576	0.7461	0.7869	0.9515	0.8109
Rhythm features with SQI filter	Yes	No	No	0.853	0.9517	0.8343	0.9692	0.8017	0.8177	0.9542	0.8132
All 17 features with SQI filter	Yes	Yes	No	0.759	0.9370	0.8430	0.9510	0.7196	0.7764	0.9543	0.8142
All 17 features with full feature-level SQI	Yes	Yes	Yes	0.763	0.9374	0.8430	0.9514	0.7214	0.7775	0.9542	0.8142
Quality features only	Yes	Yes	Yes	0.668	0.8893	0.6032	0.9319	0.5693	0.5857	0.8462	0.6134

The ablation shows that SQI filtering modestly improves a rhythm-feature logistic baseline, increasing record-level F1 from 0.8137 to 0.8177. Quality-weighted aggregation also gives a small gain for the 17-feature logistic baseline, increasing F1 from 0.7764 to 0.7775. However, simply adding SQI-related features to a linear model does not automatically improve the chosen operating point: the all-feature logistic models have lower precision and F1 than the rhythm-only logistic model, despite similar AUROC and AUPRC. This suggests that SQI is useful as a coverage and reliability mechanism, but that its best use is not necessarily raw linear feature inclusion. The full hybrid model still achieves the strongest operating-point F1 and precision among the completed p00-p02 experiments.

7.4 Neural Branch Ablation

To test which neural input stream contributes most strongly, four branch variants were retrained on the same p00-p02 SQI-accepted split: full fusion, waveform-only, spectral-only, and feature-only neural models. For each model, the decision threshold was selected on validation record-level F1 and fixed for test evaluation. This is an independent retraining ablation, so the full-fusion row is not the same checkpoint as the retained final SQI model reported later in this chapter.

This ablation shows that the waveform branch carries the strongest operating-point evidence among the retrained neural variants, with record-level F1 0.8072. The spectral-only and feature-only

Table 7.7: Neural branch ablation at record level on the p00-p02 test fold.

Neural variant	Active input branches	Threshold	Epochs run	Best epoch	Accuracy	Sensitivity	Specificity	Precision	F1	AUROC	AUPRC
Full fusion retrain	waveform + STFT + features	0.596	20	10	0.9451	0.8547	0.9586	0.7548	0.8016	0.9298	0.7930
Waveform-only	waveform	0.493	28	18	0.9472	0.8517	0.9614	0.7670	0.8072	0.9327	0.8217
Spectral-only	STFT	0.531	30	22	0.9345	0.8488	0.9473	0.7062	0.7710	0.9475	0.8128
Feature-only neural	17 features	0.606	30	20	0.9306	0.8735	0.9391	0.6814	0.7656	0.9447	0.8135

neural models have similar threshold-independent ranking performance, but lower precision and F1 at their validation-selected thresholds. The independently retrained full-fusion model does not outperform the waveform-only model in this run, even though the retained final SQI checkpoint reported later in this chapter remains the strongest overall operating point. This means the branch ablation supports the importance of learned waveform morphology, but it does not prove a stable additive gain from all three branches under the current training setup. Multiple random seeds and further fusion tuning would be needed before claiming that gated fusion is consistently superior to a waveform-only neural model.

7.5 Final SQI Validation and Test Results

The final SQI model stopped early after 18 epochs, with the best validation epoch at epoch 8. The validation-selected threshold was 0.713.

Table 7.8: Record-level validation and test results for the final SQI model.

Split	Accuracy	Sensitivity	Specificity	Precision	F1	AUROC	AUPRC
Validation	0.9782	0.8736	0.9975	0.9849	0.9259	0.9823	0.9590
Test	0.9536	0.8241	0.9729	0.8194	0.8217	0.9272	0.8365

Bootstrap 95% confidence intervals were computed on the held-out test record-event predictions. Two intervals are reported: ordinary record-event bootstrap and a more conservative subject-clustered bootstrap, where subjects rather than individual record-events are resampled.

Table 7.9: Bootstrap 95% confidence intervals for final record-level test performance.

Metric	Test value	Record-event 95% CI	Subject-clustered 95% CI
Accuracy	0.9536	0.9481–0.9593	0.9158–0.9859
Sensitivity	0.8241	0.7935–0.8526	0.6652–0.9530
Specificity	0.9729	0.9685–0.9777	0.9433–0.9917
Precision	0.8194	0.7918–0.8488	0.6683–0.9391
F1	0.8217	0.7988–0.8438	0.6885–0.9310
AUROC	0.9272	0.9114–0.9410	0.8471–0.9959
AUPRC	0.8365	0.8080–0.8619	0.6726–0.9702

The subject-clustered intervals are much wider because the p00-p02 test set contains only 103 subjects. They are therefore a useful reminder that many record-events can still represent a smaller number of individuals.

There is a visible validation-test gap, especially at record level where F1 decreases from 0.9259 to 0.8217 and AUPRC decreases from 0.9590 to 0.8365. This gap is important to report because the test fold has lower AF prevalence and a different held-out record-event composition from the validation fold. The final result should therefore be interpreted as promising performance on the held-out p00-p02 test fold, not as proof of uniform generalisation across the full MIMIC-III-Ext-PPG release.

Table 7.10: Segment-level validation and test results for the final SQI model.

Split	Accuracy	Sensitivity	Specificity	Precision	F1	AUROC	AUPRC
Validation	0.9744	0.8329	0.9923	0.9316	0.8795	0.9698	0.9122
Test	0.9526	0.8843	0.9594	0.6841	0.7714	0.9525	0.7947

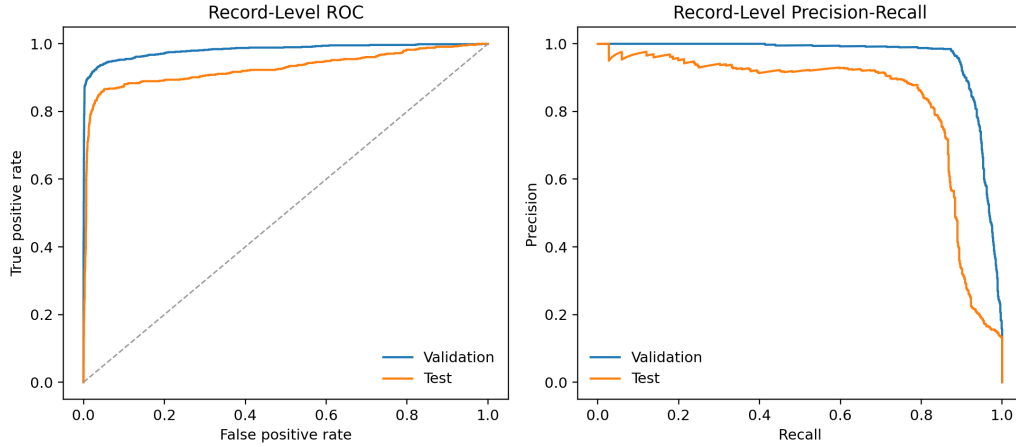


Figure 7.1: Validation and test record-level ROC and precision-recall curves for the final SQI model.

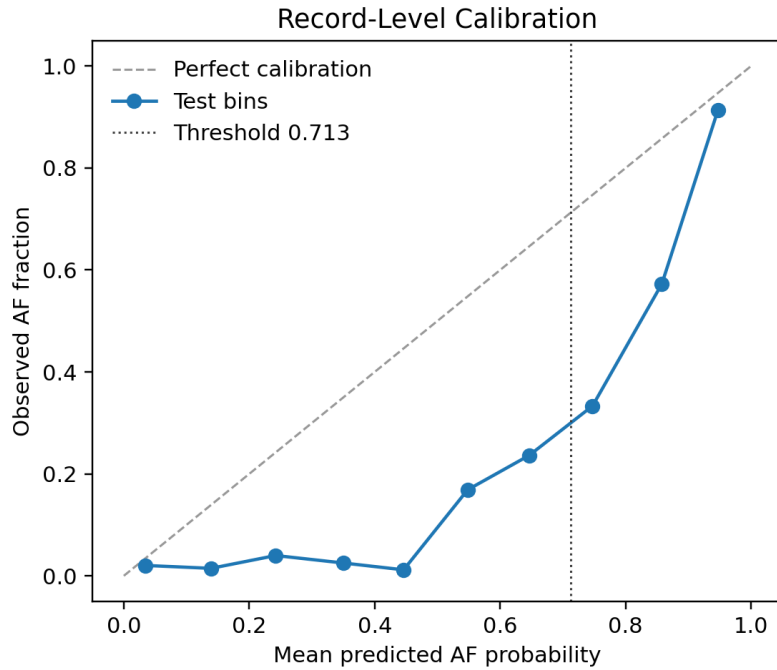


Figure 7.2: Test record-level calibration curve using ten probability bins. The vertical dotted line marks the validation-selected threshold of 0.713.

7.6 Confusion Matrices

Table 7.11: Record-level test confusion matrix at threshold 0.713.

	Predicted SR	Predicted AF
True SR	4488	125
True AF	121	567

Table 7.12: Segment-level test confusion matrix at threshold 0.713.

	Predicted SR	Predicted AF
True SR	65,090	2,757
True AF	781	5,971

The record-level confusion matrix explains the final positive-class metrics:

$$\text{Precision} = \frac{567}{567 + 125} = 0.8194, \quad \text{Sensitivity} = \frac{567}{567 + 121} = 0.8241,$$

$$\text{Specificity} = \frac{4488}{4488 + 125} = 0.9729, \quad \text{F1} = 0.8217.$$

7.7 Threshold Analysis

The primary threshold is selected on validation data. A test-set threshold sweep is included only to understand the trade-off between sensitivity and precision.

Table 7.13: Record-level threshold analysis on the test fold.

Threshold	Source	Accuracy	Sensitivity	Specificity	Precision	F1	FP	FN	TP
0.713	Validation-selected	0.9536	0.8241	0.9729	0.8194	0.8217	125	121	567
0.834	Test-best analysis only	0.9587	0.7907	0.9837	0.8788	0.8324	75	144	544

The higher test-best threshold would reduce false positives from 125 to 75 and increase precision from 0.8194 to 0.8788, but it would also increase false negatives from 121 to 144 and reduce sensitivity from 0.8241 to 0.7907. This confirms that the operating point should be chosen according to intended use.

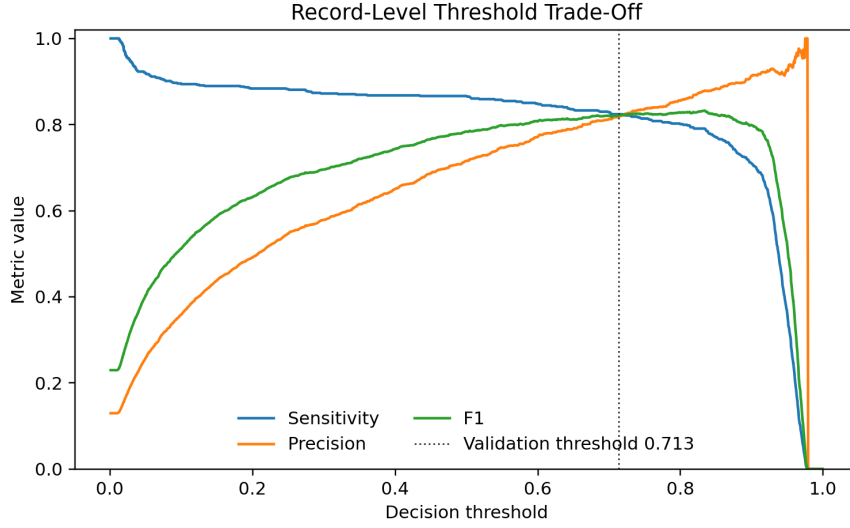


Figure 7.3: Record-level threshold trade-off for the final SQI model. The vertical reference line marks the validation-selected threshold of 0.713.

7.8 Aggregation Ablation

Because the final decision is made at record-event level, aggregation is part of the model design. Using saved segment probabilities from the final SQI model, several aggregation rules were compared. For each rule, the threshold was selected on validation record-level F1 and then fixed on the test set.

Table 7.14: Aggregation ablation using saved segment probabilities from the final SQI model.

Aggregation method	Validation-selected threshold	Test accuracy	Test sensitivity	Test specificity	Test precision	Test F1
Quality-weighted mean	0.713	0.9536	0.8241	0.9729	0.8194	0.8217
Unweighted mean	0.710	0.9532	0.8241	0.9725	0.8170	0.8205
Median	0.675	0.9493	0.8401	0.9655	0.7843	0.8112
Maximum probability	0.897	0.9402	0.8183	0.9584	0.7457	0.7803
Top-5 mean	0.772	0.9419	0.8358	0.9577	0.7468	0.7888

Quality-weighted averaging gives the best F1 among these aggregation choices, although the gain over a simple mean is small. The larger difference is between averaging-based aggregation and maximum or top- k pooling. Max-style aggregation is more vulnerable to isolated high-probability false-positive windows, which is why it has lower precision and F1.

7.9 Record-Level Error and Reliability Analysis

A lightweight error analysis was performed using `test_record_predictions.csv` at the validation-selected threshold of 0.713. This does not replace a full ablation study, but it helps identify where the retained model is less reliable.

Table 7.15: Error profile by record-level mean-quality quartile.

Quality quartile	Records	AF records	FP	FN	Sensitivity	Specificity	Precision	F1
0.780–0.918	1326	121	14	64	0.4711	0.9884	0.8028	0.5938
0.918–0.934	1325	92	27	21	0.7717	0.9781	0.7245	0.7474
0.934–0.945	1325	192	38	11	0.9427	0.9665	0.8265	0.8808
0.945–0.973	1325	283	46	25	0.9117	0.9559	0.8487	0.8790

Table 7.16: Error profile by accepted segment count per record-event group.

Accepted segments per group	Records	AF records	FP	FN	Sensitivity	Specificity	Precision	F1
1	401	129	13	41	0.6822	0.9522	0.8713	0.7652
2	258	56	2	20	0.6429	0.9901	0.9474	0.7660
3–5	624	97	13	12	0.8763	0.9753	0.8673	0.8718
6–10	869	141	14	14	0.9007	0.9808	0.9007	0.9007
11–20	1594	162	31	28	0.8272	0.9784	0.8121	0.8196
21+	1555	103	52	6	0.9417	0.9642	0.6510	0.7698

Table 7.17: Summary of mean quality, mean segment count, and mean probability by error outcome.

Outcome	Mean quality	Mean segment count	Mean probability
True positive	0.9404	10.47	0.9329
False positive	0.9368	15.45	0.8603
False negative	0.9165	6.73	0.2071
True negative	0.9279	14.69	0.0842

Representative Test-Set PPG Waveforms by Prediction Outcome

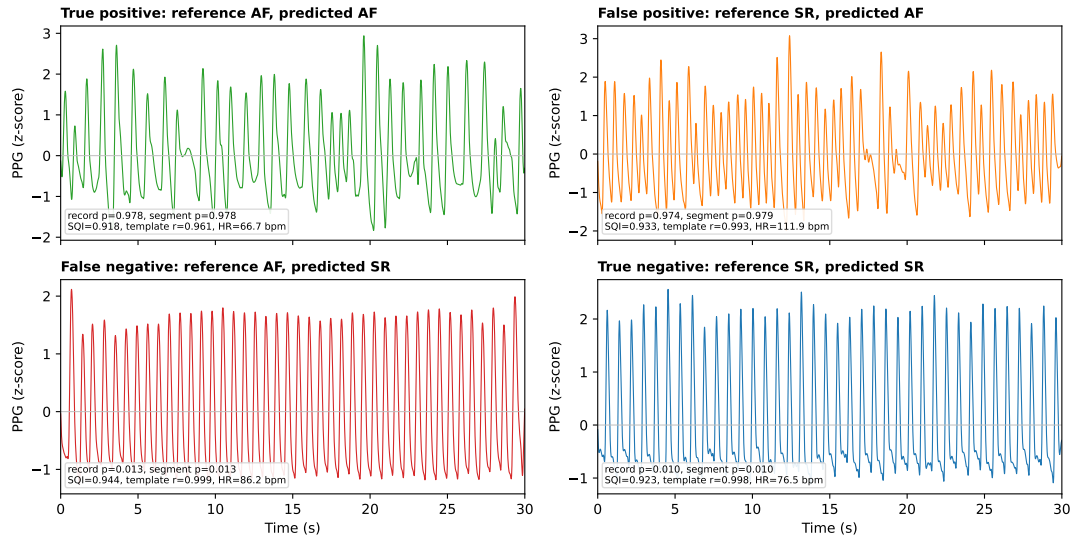


Figure 7.4: Representative test-set PPG waveform examples by prediction outcome. The panels illustrate how true positives, false positives, false negatives, and true negatives can differ in regularity, morphology stability, and apparent noise burden.

False negatives have the lowest mean quality and fewest accepted segments, suggesting that weak or sparse accepted evidence is a major failure mode. False positives have relatively high mean probability and more accepted segments, indicating that some SR record-events show persistent AF-like evidence rather than isolated single-window errors. Figure 7.4 provides representative waveform examples for these outcome groups. These analyses are descriptive and partly confounded by case mix and AF prevalence, so they should be treated as guidance for future ablation and error review rather than as causal proof.

7.10 Operating-Point Interpretation

The final model is best understood through its operating point rather than through AUROC alone. AUROC measures ranking over all thresholds, but a deployed screening system must eventually choose when to alert. At the validation-selected threshold, the final model favours high specificity and good precision while retaining useful sensitivity. This is appropriate for a user-facing or semi-automated alerting context because false positives have practical cost. Repeated false alerts could increase anxiety, increase clinician workload, and reduce trust in later alerts.

The threshold analysis shows that a higher threshold would reduce false positives and increase precision, but would miss more AF record-event groups. This is not a failure of the model; it is the normal trade-off between sensitivity and precision. A clinician-facing review tool might prefer a lower threshold to capture more possible AF events, while a consumer-style notification might prefer a higher threshold to reduce unnecessary alerts. The model should therefore be reported with its selected operating point and with a threshold-sweep analysis, not with a single claim that the model is simply “accurate”.

7.11 Calibration and Probability Meaning

The calibration curve is important because the model outputs probabilities, but a neural probability is not automatically a calibrated clinical risk. The test calibration curve suggests that predicted probabilities should be treated primarily as ranking and thresholding scores. In several probability bins, the observed AF fraction is lower than the mean predicted probability, indicating over-confident or miscalibrated probabilities in those ranges. This does not prevent the model from being useful as a classifier, because the AUROC, AUPRC, F1, and confusion matrix are evaluated at the selected threshold. However, it means the raw probability should not be displayed to a user as a literal probability of having AF without further calibration.

If the system were developed for deployment, the next step would be to calibrate probabilities on an external validation set using methods such as temperature scaling, Platt scaling, or isotonic regression, and then test whether calibration remains stable across devices and subgroups [28]. A calibrated probability could support risk communication and clinician review. An uncalibrated probability should be used more cautiously, for example as an internal score behind an AF-like pattern notification.

7.12 Statistical Uncertainty and Dependence

The report includes both record-event bootstrap and subject-clustered bootstrap intervals. This is an important addition because record-events are not fully independent when multiple events come from the same subject. Ordinary bootstrapping can make uncertainty appear smaller by resampling many correlated record-events. Subject-clustered bootstrapping is more conservative because it resamples subjects and preserves the dependence among their record-events.

The wider subject-clustered intervals should be interpreted as a caution rather than a weakness. They show that the numerical point estimate is promising, but that patient-level generalisation is less certain than record-event counts alone suggest. This is especially relevant because the p00–p02 test set contains many record-event groups but a smaller number of unique subjects. For a medical screening problem, subject-level uncertainty is more meaningful than window-level uncertainty.

7.13 Why Segment-Level Metrics Remain Secondary

Segment-level results are useful for debugging but should not be the headline claim. A single 30-second PPG window can be corrupted by motion, low perfusion, or transient poor contact. Even if it is accepted by SQI, it may contain too few beats or ambiguous morphology. Segment-level precision is therefore more sensitive to isolated false positives, especially in an imbalanced test split. The final report correctly treats segment-level metrics as diagnostic analysis and record-level metrics as the primary screening result.

This is also aligned with the behaviour of practical wearable algorithms. A user should not receive an AF alert because one short window produced a high probability. The system should require repeated or sustained evidence, and it should take signal quality into account. The record-event aggregation used here is a retrospective approximation of that principle. A future streaming system could extend it with temporal hysteresis, minimum-evidence duration, or repeated-window confirmation before alerting.

7.14 Suggested Additional Evaluation Table

A concise way to strengthen the evaluation chapter is to add a table that separates three kinds of validity evidence. This prevents the report from sounding as if one held-out test score proves everything.

Validity question	Evidence already available	Remaining gap
Can the model distinguish AF from SR in accepted p00–p02 windows?	Segment and record-level test metrics, AUROC, AUPRC, confusion matrices	Full-prefix and external dataset validation
Does record-level aggregation improve screening behaviour?	Aggregation ablation showing quality-weighted mean outperforms max-style pooling	Streaming alert policy with no-call coverage
Is the operating point defensible?	Validation-selected threshold, test confusion matrix, threshold sweep	Prospective threshold calibration for specific use cases
Are results robust to dependence?	Record-event and subject-clustered bootstrap intervals	Larger subject-level external validation
Can the method run near the sensor?	Embedded 17-feature model smoke test	Live C feature extraction and hardware validation

Table 7.18: Different types of evidence provided by the evaluation and the remaining validation gaps.

7.15 Integrated Interpretation of the Final Test Result

The final record-level result can be interpreted from the confusion matrix. At the selected threshold, 567 of 688 AF record-event groups are detected, while 4,488 of 4,613 SR groups are correctly rejected. This gives a balanced result for retrospective screening: sensitivity is high enough to detect most AF record-events, and specificity is high enough to limit false alerts. The precision of 0.8194 is particularly important because the test set is imbalanced; under imbalance, even a small false-positive rate can reduce precision.

The segment-level result tells a different story. Segment-level sensitivity is higher than record-level sensitivity, but precision is lower. This is consistent with the idea that many individual 30-second windows contain useful AF evidence, but isolated SR windows can still appear AF-like. Aggregation reduces the impact of those isolated windows by requiring evidence across a record-event group. Therefore, the main conclusion should emphasise record-level screening performance rather than single-window classification.

7.16 What the Ablations Do and Do Not Prove

The feature-level SQI ablation supports the usefulness of SQI filtering, but it does not fully isolate every SQI contribution in the neural model. The rhythm-feature logistic model improves modestly when SQI filtering is applied, while simply adding SQI features to a linear model does not always improve F1. This suggests that SQI is valuable as a reliability and coverage mechanism, but that its effect depends on how the information is used. The strongest interpretation is therefore that SQI improves the overall screening pipeline, not that every individual SQI feature monotonically improves every classifier.

The neural branch ablation shows that learned waveform morphology is highly informative. The waveform-only retraining achieved stronger operating-point F1 than the independently retrained full-fusion variant. This means the thesis should avoid saying that the multi-branch fusion architecture is proven superior in a strict ablation sense. A more defensible statement is that the retained final hybrid checkpoint achieved the strongest overall operating point among the completed experiments, while the branch ablation indicates that the time-domain morphology branch is the most consistently valuable neural input stream.

The aggregation ablation is more directly supportive. Maximum probability and top-k pooling perform worse than averaging-based approaches, which is consistent with the expected failure mode of isolated high-probability false-positive windows. Quality-weighted averaging gives the best F1, although the gain over unweighted mean is small. The practical lesson is that aggregation should be conservative and evidence-averaging rather than alerting on a single extreme window.

7.17 Failure Modes Suggested by the Error Analysis

The record-level error analysis suggests two different failure modes. False negatives have lower mean quality and fewer accepted segments, which suggests that weak or sparse evidence can cause AF record-events to be missed. In a streaming deployment, this could happen when AF occurs during poor sensor contact, low perfusion, or motion. Repeated monitoring might eventually detect later windows, but this cannot be assumed without prospective longitudinal validation.

False positives have relatively high mean probability and more accepted segments. This is a different problem: the model is not merely reacting to one noisy window, but may be seeing persistent AF-like evidence in some SR-labelled groups. Possible explanations include ectopy, atrial flutter-like or irregular non-AF rhythms excluded from the binary label set, residual label timing mismatch, or PPG artefacts that pass the SQI gate. This is the main argument for future multi-rhythm classification rather than only AF-versus-SR detection.

The representative waveform figure is useful because it makes these failure modes concrete. The false-negative example appears more regular than the true-positive example, while the false-positive example shows persistent irregularity despite an SR reference label. This illustrates why PPG AF detection cannot rely on irregularity alone: some AF can appear less irregular over short windows, and some non-AF or artefactual windows can appear irregular.

7.18 Comparison With Wearable Screening Studies

The final results should not be compared numerically with Apple Heart Study or Fitbit Heart Study as if all studies used the same endpoint. Those studies evaluated wearable notification algorithms in large participant cohorts with ECG-patch follow-up, while this thesis evaluates retrospective ICU-derived record-event classification from MIMIC-III-Ext-PPG. Nevertheless, the design logic is similar in two important ways. First, both large wearable studies and this thesis use conservative evidence accumulation rather than one-window alerts. Second, both maintain a distinction between PPG-based suspicion and ECG-based confirmation [11, 13].

This comparison strengthens the thesis because it shows that the project is not trying to solve an artificial classification problem. The record-level aggregation and SQI gate mirror the practical logic used by real-world wearable screening systems: ignore unreliable windows, accumulate evidence, then issue a cautious notification or screening decision.

Chapter 8

Embedded Edge AI Deployment

After evaluating the retrospective PPG models, the project also investigates whether the feature-based part of the pipeline can be connected to an embedded inference workflow. This chapter is placed after the performance results because the deployment work is a downstream implementation step rather than the primary source of model-performance evidence.

8.1 Deployment Rationale and Scope

In this project, Edge AI means running the trained inference model on or close to the sensing device, rather than sending all raw data to a remote cloud server for analysis. The “edge” is the local embedded device at the boundary between the physical world and the digital system: in this project, a wearable-class nRF9160/Zephyr device connected to the PPG monitoring workflow. This does not mean that model training is performed on the device. Training and model development are performed retrospectively in Python, while the embedded component tests whether a compact trained model can execute locally.

The Edge AI component is included because PPG screening is naturally tied to physical sensing hardware. A practical wearable system would collect PPG at the wrist or another body location, decide whether the signal is usable, and then either provide a local screening output or forward only selected information for review. In that setting, sending every raw waveform window to a cloud server is not always desirable. Connectivity may be intermittent, power is limited, latency matters for user feedback, and physiological data are sensitive. Bringing inference closer to the source is therefore not only an implementation detail; it is part of the practical feasibility of PPG-based monitoring.

This is also relevant to the industry-facing context of the project. The target direction is closer to a specific-purpose patient-monitoring device than to a general-purpose smartwatch application. Such devices need a clear path from retrospective model development to constrained embedded inference, even if the first embedded version is only a compact proof of concept.

The scope of this section is deliberately limited. The embedded work does not claim that the full waveform/STFT/feature fusion model has been reproduced on the device, and it does not provide clinical hardware validation. Instead, it demonstrates that the 17-feature representation can be packaged into a Nordic Edge AI Lab model, linked into a Zephyr firmware project, initialised through the generated runtime API [29], and exercised with holdout feature vectors. This provides evidence for deployment-path feasibility while keeping the main model-performance claim in Chapter 7.

8.2 Firmware Target

The embedded component uses the firmware project `E2_V3_NCS2.3.0_60115_Copy`, a Zephyr/nRF Connect SDK application targeting an nRF9160-class device. The firmware includes modules for display, GPS, UART/BLE, power management, IMU support, and PPG support. The Edge AI integration is included through:

```
src/test2_92249_v3
src/test2_92249_v3/nrf_edgeai_generated
src/test2_92249_v3/nrf_edgeai/lib/libnrf_edgeai_cortex-m33.a
src/test2_92249_v3/edgeai_smoke/edgeai_smoke.c
```

The top-level firmware CMake file adds the module with:

```
add_subdirectory(src/test2_92249_v3)
```

8.3 Nordic Edge AI Lab Model

The generated Nordic Edge AI Lab model has solution ID 92249. The generated header exposes:

```
nrf_edgeai_t* nrf_edgeai_user_model_92249(void);
uint32_t nrf_edgeai_user_model_neuton_size_92249(void);
```

The generated code aliases this to the standard `nrf_edgeai_user_model()` API. The model is a Neuton CPU model, not an Axon/NPU model.

Table 8.1: Compact Nordic Edge AI Lab model properties.

Property	Value
Input features	17 float32 values
Input window size	1
Output classes	2
Internal model neurons	4
Model weights	16
Model quantisation	32 bits
Model flash usage reported by runner	133 bytes

This deployment model is deliberately smaller than the main hybrid model. It demonstrates an embedded inference path using the same feature representation, while the larger waveform/STFT/feature fusion model remains the main retrospective performance model.

8.4 Embedded Smoke Test

The firmware includes an embedded smoke-test thread. At startup, the thread waits two seconds, obtains the Edge AI model pointer, checks the expected input count, calls `nrf_edgeai_init()`, and then feeds holdout samples using:

```
nrf_edgeai_feed_inputs(p_edgeai, sample_buf, expected_inputs);
nrf_edgeai_run_inference(p_edgeai);
```

The holdout CSV contains 2,249 samples with 17 feature columns and one binary label column. The build system converts the CSV into a C header using `csv_to_edgeai_header.py`, storing the data as const arrays in flash. The smoke test reports matched samples, total samples, accuracy, mismatch count, and the first mismatched rows through Zephyr logging.

The same generated model was also checked with Nordic’s desktop inference runner on the 2,249-row holdout CSV. The runner reported accuracy 0.9582 and balanced accuracy 0.9478. This confirms that the generated model and feature order behave consistently in the host-side runtime before being exercised through the Zephyr smoke-test path.

This checks that the generated model, feature order, runtime library, CMake integration, and firmware inference calls are consistent.

8.5 Deployment Result and Interpretation

The embedded deployment component successfully integrates a compact 17-feature Nordic Edge AI Lab model into the Zephyr firmware project. The current result is a firmware-level smoke-test

harness rather than a full hardware validation study. It confirms that:

- the generated Edge AI model is available through the expected C API,
- the model expects 17 float32 input features,
- the holdout dataset can be converted into a flash-resident C header,
- the desktop runner obtains 0.9582 accuracy and 0.9478 balanced accuracy on the same 2,249 holdout rows,
- the firmware can initialise the runtime, feed input vectors, execute inference, and compare predicted classes against labels.

This supports the feasibility of on-device inference, but the deployed compact model should be presented separately from the main hybrid model. The value of the result is therefore not that it matches the retrospective hybrid model’s performance, but that it shows a credible path from the PPG feature pipeline to an embedded runtime. In a larger project, this would be followed by C implementation of feature extraction, Python-versus-firmware feature equivalence testing, live or replayed PPG inference on the device, and ECG-confirmed hardware validation.

8.6 Deployment Gap Analysis

The deployment result should be evaluated against what it proves and what it does not prove. It proves that a compact model based on the 17-feature representation can be generated, linked into the Zephyr firmware project, initialised through the expected API, and exercised using holdout feature vectors. This is valuable because it demonstrates that the feature-level part of the pipeline can cross the boundary from Python experimentation to embedded inference.

It does not prove that the full offline hybrid network runs on the target device. The full model includes waveform convolutions, a Transformer component, STFT processing, and gated fusion. Running that model on-device would require a separate optimisation study involving model compression, quantisation, memory profiling, operator support, and latency measurement. It also does not prove that live PPG can be processed end-to-end in firmware, because the smoke test uses precomputed features rather than C-derived features from the raw sensor stream.

A concise way to present the deployment contribution is therefore: the project demonstrates embedded inference integration for the compact feature model, not embedded reproduction of the full offline research model. This is still a meaningful MEng contribution because it connects the algorithmic representation to a realistic firmware pathway and identifies the exact remaining steps needed for end-to-end deployment.

8.7 Production-Oriented Verification Plan

A future deployment study should progress through four verification levels. The first level is runtime verification, which this project has already started: the model can be initialised, fed inputs, and used to produce output classes. The second level is feature-equivalence verification: the same raw PPG segment should produce numerically similar 17-feature vectors in Python and firmware. This requires a C implementation of filtering, peak detection, SQI checks, and feature extraction, followed by tolerance-based comparison against the Python pipeline.

The third level is replay verification. Recorded PPG segments should be streamed or replayed through the firmware path, including signal buffering, feature extraction, inference, no-call logic, and logging. This would test whether the firmware handles realistic timing and memory behaviour rather than only static arrays. The fourth level is prospective hardware validation with synchronised ECG confirmation. Only this final level can support claims about live device performance, alert burden, no-call coverage, and user-state robustness.

Verification level	What it tests	Current status
Runtime smoke test	Generated model API, input count, inference call, output comparison	Demonstrated for 17-feature model
Feature equivalence	Python and C pipelines produce matching features from the same raw PPG	Future work
Replay test	Firmware processes recorded PPG in a realistic streaming workflow	Future work
Prospective validation	Live sensor performance with ECG-confirmed labels and alert/no-call analysis	Future work

Table 8.2: Suggested staged verification plan for moving from embedded smoke test to live deployment.

8.8 Why Edge Inference Is Relevant for PPG Screening

Edge inference is relevant here because PPG is generated continuously at the sensor. Sending every raw waveform to a remote server may be undesirable because of connectivity, latency, power, and privacy constraints. A local model can decide whether the signal is usable, compress the evidence into features or probabilities, and transmit only selected summaries when needed. This does not remove the need for clinical confirmation, but it can make the monitoring workflow more practical.

The Zephyr ecosystem provides examples of running TensorFlow Lite for Microcontrollers in embedded applications [16], and Arm CMSIS-NN is designed to improve neural-network performance and memory footprint on Cortex-M processors [17, 18]. Although this project uses a Nordic Edge AI Lab model rather than a full TFLite Micro deployment, these references provide broader context: embedded machine-learning deployment is a realistic engineering direction, but it requires careful attention to model size, operator support, memory, latency, and verification.

Chapter 9

Discussion

9.1 Main Finding

The key finding is that SQI-aware PPG processing can support promising retrospective record-level AF detection on the evaluated ICU-derived p00–p02 subset. The final MIMIC-III-Ext-PPG result has high specificity (0.9729), good precision (0.8194), and balanced F1 (0.8217) at the validation-selected threshold. This is encouraging for screening because high specificity reduces unnecessary alerts, while sensitivity above 0.82 indicates that most AF record-event groups are detected.

The result also shows the limits of small controlled baselines. MIMIC PERform AF achieved perfect record-level test performance, but that result is based on only five test records. It is useful as a sanity check, not as the main claim. The larger p00–p02 subset reveals the more realistic challenges: imbalance, heterogeneous ICU physiology, and sensitivity–specificity trade-offs.

9.2 Segment-Level and Record-Level Metrics Tell Different Stories

Segment-level test sensitivity is higher than record-level sensitivity (0.8843 versus 0.8241), but segment-level precision is much lower (0.6841 versus 0.8194). This occurs because individual 30-second windows are noisy and the segment-level AF prevalence in the test split is only 9.05%. Even a modest number of false-positive SR windows can reduce segment-level precision.

Record-level aggregation improves practical interpretability. By averaging probabilities across a record-event group and weighting by quality, isolated false-positive windows have less influence. This is closer to how a screening system would operate, because alerts should be based on repeated or episode-level evidence rather than a single local window.

9.3 Interpretation of Precision and Specificity

At the selected threshold, record-level specificity is high. The model correctly rejects 4,488 of 4,613 SR record-event groups. This is important because false-positive AF alerts can create unnecessary anxiety and clinical workload.

The cost is that 121 of 688 AF record-event groups are missed. Whether this operating point is acceptable depends on deployment context. For user-facing alerts, higher precision and specificity may be preferred. For clinician-facing retrospective review, a lower threshold may be acceptable to increase sensitivity.

9.4 Role of SQI

SQI is built into three stages: it filters unreliable segments before training and evaluation, it contributes explicit quality features to the model, and it weights both the loss and record-level

aggregation. This is important because AF-like irregularity can be produced by signal artefact. The model should not be rewarded for detecting irregularity in segments where the pulse train itself is unreliable.

Across the completed p00–p02 development runs, the retained SQI pipeline is associated with improved precision, specificity, F1, and AUPRC compared with the initial large-scale hybrid run. However, because a full no-SQI, SQI-filtering-only, SQI-features-only, and full SQI pipeline ablation was not completed, the independent contribution of SQI should be interpreted as plausible and supported by the observed pipeline behaviour, but not causally isolated.

The feature-level ablation adds modest but consistent support for this interpretation. Applying SQI filtering to the feature-only logistic baseline increased record-level F1 from 0.8137 to 0.8177, showing that even a simpler model benefits when obviously unreliable segments are removed before aggregation. Coverage analysis also reduces one possible concern: the SQI gate did not obtain strong performance by discarding most of the data. Accepted-window coverage remained high across the evaluated splits, so SQI remains part of the reliability profile rather than a preprocessing detail that can be ignored.

9.5 Edge Deployment Interpretation

The Edge AI integration demonstrates a practical deployment pathway but does not yet reproduce the full offline model on-device. The offline model uses waveform CNNs, a Transformer, STFT processing, and gated fusion. The embedded model uses only the 17-feature vector and a compact Neuton CPU classifier. This is a reasonable engineering split: the offline model explores performance, while the embedded model demonstrates feasibility under device constraints.

The reason for including this work is that PPG AF screening is ultimately a sensing problem as well as a modelling problem. A wearable system must operate close to the signal source under constraints that are not visible in a retrospective Python experiment: limited memory, limited compute, battery constraints, intermittent connectivity, and privacy-sensitive physiological data. The firmware integration therefore tests whether the project has a credible path beyond retrospective classification. It shows that a compact version of the feature representation can be converted into an embedded inference artefact and invoked through the target runtime.

This should be interpreted as proof-of-concept evidence, not as a final deployment result. It does not yet show live PPG acquisition, C-based feature extraction, or equivalence between embedded features and the Python feature pipeline. It also does not show that the full hybrid model can fit within the present device constraints. Its contribution is narrower but still useful: it reduces the gap between the research pipeline and a future specific-purpose patient-monitoring wearable by showing that on-device inference is technically reachable.

The next deployment step is to implement the feature extraction pipeline in C, compare embedded features against the offline Python pipeline, and then test end-to-end inference on live or replayed PPG from the target hardware with ECG-confirmed labels.

9.6 Ablation and Remaining Error Analysis Gaps

The current results are strongest as an end-to-end feasibility study. The record-level error analysis reported earlier in this chapter adds useful evidence: false negatives tend to have lower mean quality and fewer accepted segments, while false positives often have persistent high-probability evidence across more segments. However, the study does not yet isolate every design choice. The most useful next analyses would be SQI ablations, branch ablations, aggregation ablations, and deeper false-positive and false-negative review under repeated-seed settings. These analyses would move the evidence from retrospective feasibility toward robustness.

Without these studies, the safest interpretation is that the method is promising on the evaluated p00–p02 subset but still needs broader validation.

9.7 Clinical Workflow Interpretation

A practical way to interpret the final system is as the first stage of a two-stage workflow. The first stage is PPG-based screening: the system observes accepted PPG windows, estimates AF-like probability, aggregates evidence across a record-event, and either produces a suspicious-rhythm output or abstains when signal quality is inadequate. The second stage is confirmation: a clinician or ECG-capable device verifies the rhythm before treatment decisions are made.

This workflow is consistent with the project results. The model achieves high specificity and good precision, which are desirable for reducing unnecessary alerts. However, it still misses some AF record-events and it is not perfectly calibrated. Therefore, a negative result should not be interpreted as ruling out AF, and a positive result should not be interpreted as a diagnosis. The appropriate message is: the PPG evidence is or is not sufficient to raise suspicion under the current threshold and signal-quality constraints.

9.8 Why Precision Matters in This Dataset

Precision is especially important in this project because the test set contains many more SR than AF record-event groups. Under such imbalance, false positives can accumulate quickly. A model with good sensitivity but poor precision might identify many AF cases, but it would also produce many unnecessary AF alerts. This would be undesirable in a user-facing device and burdensome in a clinical triage system.

The final model improves precision compared with the initial large-scale hybrid model and the feature-only logistic baseline at the selected record-level operating point. This supports the claim that the final pipeline is not merely detecting irregularity aggressively; it is operating in a more conservative region of the sensitivity–specificity trade-off. The cost is a modest reduction in sensitivity compared with some earlier runs. That trade-off should be framed as a design choice rather than hidden.

9.9 Interpretation of the Validation–Test Gap

The validation–test gap is an important analytical result. The validation F1 is substantially higher than the test F1, and the validation AUPRC is also higher than the test AUPRC. This does not necessarily indicate a modelling error. It may reflect lower AF prevalence, different subject composition, different rhythm-event distribution, or different signal-quality patterns in the test fold. However, it does show that performance is not uniform across folds.

The safest conclusion is that the final model performs well on the held-out p00–p02 test fold but still requires wider validation. Broader evaluation and external device-specific validation would test whether the chosen SQI thresholds, feature scaling, model architecture, and threshold remain stable. This is particularly important because a medical screening system must be robust across people and contexts, not only across segments.

9.10 Contribution Relative to an MEng Project

The project contribution is strongest when presented as an end-to-end engineering study. It includes data preparation at large scale, signal processing, quality assessment, feature engineering, neural modelling, operating-point selection, statistical uncertainty estimation, ablations, error analysis, and embedded integration. The main value is not a single new neural layer. It is the integration of these components into a defensible screening pipeline and the honest separation between retrospective performance and deployment feasibility.

This framing is suitable for an MEng individual project because it demonstrates technical breadth and critical judgement. The project identifies a real clinical-engineering problem, builds a complete pipeline, evaluates it on a substantial dataset, analyses failure modes, and avoids overclaiming clinical deployment. That is a stronger contribution than reporting a high accuracy number without SQI, aggregation, or deployment analysis.

Chapter 10

Limitations

This project has several limitations. The p00–p02 subset should be understood as the defined large-scale evaluation scope of this thesis rather than as a flaw in the experiment: it already contains 789,864 accepted segments and 52,419 record-event groups after SQI filtering. The limitations below therefore concern how far the results can be interpreted beyond that controlled retrospective scope.

First, the binary task uses AF versus SR. This is interpretable and appropriate for a focused screening study, but real screening systems must also handle atrial flutter, ectopy, pacing, tachycardia, bradycardia, and other rhythms that may resemble or confound AF. The current result should therefore not be interpreted as evidence for general multi-rhythm arrhythmia classification.

Second, rhythm labels in MIMIC-III-Ext-PPG are derived from charted clinical rhythm events and associated waveform windows. They are not prospective smartwatch-style annotations or per-window adjudicated ECG labels. Although the dataset follows a careful annotation strategy, label noise and timing mismatch remain possible.

Third, the SQI thresholds are hand-designed. They worked well enough for this project and coverage is high on the evaluated subset, but optimal thresholds may differ across devices, patient populations, perfusion states, movement contexts, and sensor placements. This limits how strongly the current SQI settings can be assumed to transfer without recalibration.

Fourth, SQI coverage is reported at split level, but not as a continuous monitoring coverage study. A deployment setting would need to measure how often the system can produce valid predictions over real time, during movement, rest, sleep, low perfusion, and poor-contact periods.

Fifth, the evaluation is retrospective. A deployed screening system would require prospective testing, user-state context, synchronised ECG confirmation, and evaluation of alert burden. The current model should therefore be interpreted as a retrospective screening pipeline, not as a diagnostic device.

Sixth, the ablations do not fully isolate every design choice. The feature-level SQI ablation, neural branch ablation, and aggregation ablation strengthen the analysis, but they do not replace a full repeated-seed study of SQI filtering, SQI features, quality weighting, and fusion under identical training conditions.

Seventh, subgroup performance is not fully analysed. Quality-level and segment-count error analyses are included, but this work does not yet quantify performance across age, sex, ethnicity, diagnosis group, skin tone, perfusion state, or other clinical and physiological strata.

Eighth, the embedded model is not the same as the main hybrid model. The deployment section demonstrates that a compact 17-feature model can be integrated into the Nordic/Zephyr inference path, but it does not prove that the full waveform/STFT/feature model runs on-device, that live PPG feature extraction is validated in firmware, or that clinical hardware performance has been established.

10.1 Human-Factors Risks in AF Screening

A PPG AF screening system affects users through its wording as much as through its numerical accuracy. A message such as “AF detected” could be interpreted as a diagnosis and may cause unnecessary distress. A more appropriate message would be “possible irregular rhythm detected” or “AF-like pulse pattern observed; ECG confirmation is recommended”. This wording keeps the system within a screening role and avoids implying that a PPG-only model can make treatment decisions.

The no-call state should also be communicated clearly. If the signal is rejected by SQI, the system should not silently behave as if rhythm was normal. It should indicate that the signal was not reliable enough for assessment. Otherwise, a user may falsely believe that continuous monitoring was successful when the device was frequently unable to analyse the signal.

10.2 Bias, Coverage, and Unequal Abstention

Ethical evaluation should include coverage as well as classification performance. A model may have good sensitivity and specificity on accepted windows, but if certain groups have lower PPG quality and more rejected windows, they may receive less effective monitoring. PPG quality can vary with perfusion, movement, sensor placement, age, comorbidities, oedema, and skin-sensor optical coupling. The current project reports aggregate results and quality-based error analysis, but it does not provide a full subgroup audit.

For deployment, subgroup evaluation should measure both performance and abstention. The relevant questions are: how often does the system produce valid windows for each subgroup; what is the sensitivity and precision among accepted windows; and does the no-call rate differ systematically? A fair screening system should not only be accurate when it speaks; it should also avoid being silent disproportionately for particular users.

10.3 Data Minimisation and Edge Processing

Edge inference can support data minimisation by reducing the need to transmit raw physiological waveforms. A local device can compute features, assess quality, and transmit only summaries or alerts when needed. This may reduce privacy risk compared with continuous raw-data streaming. However, edge inference does not remove governance obligations. Live deployment would still require consent, secure storage, access control, update management, audit logging, and clear explanation of what data are processed locally and what data leave the device.

In this project, the embedded smoke test uses holdout feature vectors rather than live patient data. That keeps the deployment work within a controlled engineering demonstration. A real deployment would require a separate governance plan covering live acquisition, retention period, security, user consent, clinical escalation, and failure handling.

10.4 Regulatory Boundary

The current system should be described as research software and a screening prototype. It should not be described as a medical device ready for clinical use. Moving toward clinical deployment would require formal risk management, verification and validation, prospective testing, cybersecurity assessment, usability testing, and compliance with applicable medical-device regulations. The thesis does not need to solve those regulatory steps, but it should acknowledge them so that the project does not appear to overclaim.

Chapter 11

Future Work

Future work should prioritise the following directions.

1. Broader validation across the remaining MIMIC-III-Ext-PPG prefixes and external or device-specific PPG datasets, to test whether the p00–p02 findings remain stable under wider data composition and acquisition conditions.
2. Full neural SQI ablation studies, especially no-SQI, SQI-feature-only, SQI-filter-only, and full-SQI comparisons, plus repeated-seed branch ablations to test whether the observed waveform-only strength is stable.
3. Multi-class rhythm classification, including atrial flutter, ectopy, pacing, AV block, tachycardia, and bradycardia, so that AF-like confounders are evaluated directly rather than excluded from the binary task.
4. Deeper error analysis of false positives and false negatives using waveform examples, rhythm context, AF prevalence, signal quality, and low-support record-event groups.
5. Soft SQI modelling, where lower-quality segments are down-weighted or assigned uncertainty rather than always rejected.
6. Learned SQI models trained to distinguish motion artefact, poor contact, and low perfusion from physiologic rhythm irregularity.
7. Threshold calibration for different operating modes, such as high-sensitivity clinician review and high-specificity user-facing alerts.
8. Domain-adaptation methods to reduce distribution-specific performance variation across broader datasets and acquisition settings.
9. Demographic and clinical subgroup analysis to assess fairness, robustness, and coverage across patient groups.
10. Embedded feature extraction in C so that the Nordic firmware can process live PPG rather than precomputed feature vectors.
11. Prospective device-specific validation with synchronised ECG confirmation, including alert burden, no-call coverage, and user-state analysis.

Chapter 12

Conclusion

This project developed a PPG-only AF screening pipeline combining signal processing, SQI filtering, hybrid neural classification, record-level aggregation, and embedded Edge AI deployment feasibility work. The strongest evidence from this project is not that PPG can diagnose AF, nor that gated fusion is universally superior, but that an SQI-aware record-level PPG pipeline can produce promising retrospective AF-versus-SR screening performance on the evaluated p00–p02 subset. MIMIC PERform AF provided a small controlled sanity check, while MIMIC-III-Ext-PPG p00–p02 provided the main large-scale subset evaluation.

The final SQI model achieved record-level test accuracy 0.9536, sensitivity 0.8241, specificity 0.9729, precision 0.8194, F1 0.8217, AUROC 0.9272, and AUPRC 0.8365. The record-event bootstrap interval for F1 was 0.7988–0.8438, and SQI acceptance remained high on the evaluated subset. The final hybrid model also outperformed a 17-feature logistic baseline in record-level F1 and precision, while aggregation analysis showed that quality-weighted averaging was slightly stronger than a simple mean and more reliable than max-style pooling. A retrained neural branch ablation found that the waveform-only branch was the strongest retrained branch variant, so the safest architectural conclusion is that learned PPG morphology is highly informative, while the incremental value of gated multi-branch fusion still needs repeated-seed validation. Taken together, these results support retrospective feasibility of SQI-aware PPG AF screening on the evaluated p00–p02 subset and show that record-level aggregation provides a more useful screening summary than isolated segment decisions.

The project also demonstrates a deployment pathway through a compact Nordic Edge AI model integrated into Zephyr firmware and exercised by a smoke-test harness. This embedded component is a compact feature-based feasibility demonstration, not an on-device reproduction of the full offline hybrid model. The final system should be understood as a retrospective screening pipeline: promising for AF risk detection from PPG, but requiring confirmatory ECG, full-prefix evaluation, hardware validation, prospective validation, and further robustness analysis before clinical use.

Appendix A

Supplementary Technical Notes

A.1 Pipeline Rationale

The final system uses a modular pipeline because PPG AF screening is both a classification problem and a signal-reliability problem. Filtering and normalisation reduce amplitude and baseline variability. Peak detection provides pulse timing. SQI rejects windows where the pulse train is unlikely to be reliable. The 17-feature vector captures rhythm variability, morphology consistency, and quality evidence. The neural model combines these features with waveform and spectral branches, and record-level aggregation converts noisy 30-second predictions into a more stable record-event decision.

The 30-second window length is retained because it matches the MIMIC-III-Ext-PPG segmentation and provides enough beats for interval-variability features while limiting the chance that a long window spans multiple rhythm or noise states. At deployment time, the same logic could be adapted to overlapping or streaming windows, but that is outside the current retrospective evaluation.

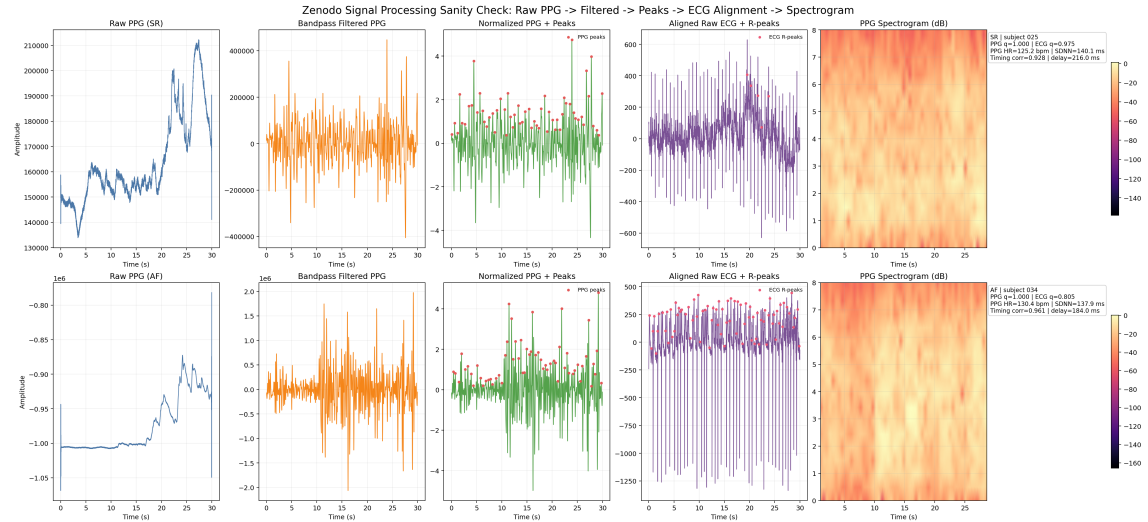


Figure A.1: Supplementary preprocessing sanity-check examples from an auxiliary annotated Zenodo PPG/ECG set used during early pipeline debugging. The panels illustrate waveform inspection, band-pass filtering, peak detection, ECG–PPG alignment, and spectrogram generation.

A.2 Components Considered but Not Retained as Main Results

Earlier project planning considered synthetic pre-training and IMU-aware motion features. These are not used as headline results because the final measured experiment is based on real MIMIC-III-

Ext-PPG p00–p02 data and PPG-derived SQI. Synthetic data may still be useful for stress testing, rare rhythm augmentation, or pre-training in future work. IMU features may improve free-living wearable robustness, but they are not part of the ICU-derived p00–p02 experiment reported here.

The binary AF-versus-SR setup is also a deliberate first step rather than a complete rhythm classifier. It isolates the central PPG AF screening question, but it does not test atrial flutter, ectopy, pacing, bradycardia, tachycardia, or other rhythm confounders. The preprocessing, SQI, feature extraction, aggregation, and embedded integration work are intended to be reusable for a future multi-class rhythm model.

A.3 Offline and Embedded Model Separation

The offline model and the embedded model are related but not equivalent. The offline model uses waveform samples, STFT evidence, and 17 hand-crafted features. The embedded Nordic Edge AI model uses only the 17-feature vector and is exercised through a Zephyr smoke-test harness. The embedded section therefore supports integration feasibility, not performance parity with the offline hybrid model.

The next deployment step would be to implement the feature extraction pipeline in C, compare embedded features against the Python pipeline, and evaluate live or replayed PPG on the target hardware with ECG-confirmed labels.

A.4 Reproducibility Notes

The most important reproducibility constraints are preserving the metadata fold split, preserving SQI preprocessing, and using the validation-selected record-level threshold rather than a test-optimised threshold. Changing the accepted segment set, the fold split, or the threshold-selection rule would change the reported operating point.

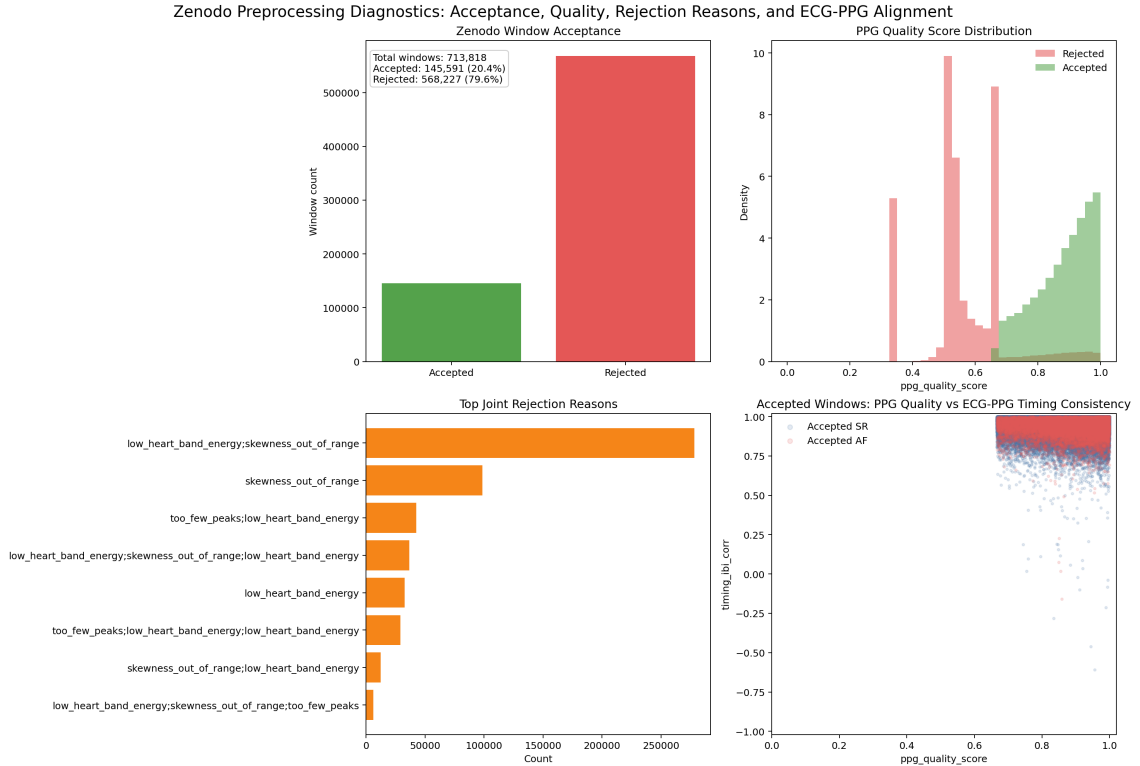


Figure A.2: Supplementary preprocessing diagnostics from the same auxiliary Zenodo sanity-check workflow. The figure summarises acceptance rates, quality-score distributions, dominant rejection patterns, and ECG–PPG timing consistency as implementation evidence rather than as a headline experimental result.

Appendix B

Ethical Considerations and Data Governance

MIMIC-III-Ext-PPG is derived entirely from MIMIC-III [9, 10, 30, 31]. According to the dataset documentation, MIMIC-III was released under Institutional Review Board oversight at Beth Israel Deaconess Medical Center and the Massachusetts Institute of Technology, with HIPAA-compliant de-identification. The extended PPG resource does not introduce new patient contact or new identifiable data; it reprocesses and curates existing MIMIC-III waveform and clinical records.

The files are credentialed PhysioNet health data. Access requires signing the appropriate data use agreement and completing required training. The project therefore treats the data as sensitive clinical research data even though they are de-identified. Results are reported only in aggregate, and no attempt is made to re-identify patients.

B.1 Screening Versus Diagnosis

The most important ethical boundary is the distinction between screening and diagnosis [1–3]. A PPG-based model can identify pulse patterns consistent with AF, but it cannot directly observe atrial electrical activity. It therefore should not be used as the sole basis for anticoagulation, cardioversion decisions, or other clinical interventions. The appropriate interpretation is that a positive PPG prediction indicates elevated suspicion and should trigger confirmatory ECG or clinician review.

This distinction should also be visible in any deployed user interface. A system should avoid language such as “you have AF” and instead use wording such as “possible irregular rhythm detected” or “AF-like pattern detected, seek confirmation.” The model’s probability output and SQI state can support this by communicating uncertainty. Low-quality segments should produce an unreliable-signal state rather than a confident AF/SR label.

B.2 False Positive Burden

False positives matter because AF alerts can cause anxiety and may increase clinical workload. In a consumer wearable setting, repeated false alerts can reduce trust and cause users to ignore later warnings. In a clinical setting, false positives can lead to unnecessary review, ECG confirmation, or monitoring. For this reason, precision and specificity are central to the evaluation.

The final SQI operating point is relatively conservative, with high record-level specificity and good precision. This makes sense for an alerting system where false positives have practical cost. However, no threshold is universally correct. A system intended for silent retrospective triage might accept more false positives to improve sensitivity. A user-facing smartwatch alert might require stricter specificity. Threshold calibration should therefore be matched to the deployment environment.

B.3 False Negative Risk

False negatives are also ethically important because missed AF may delay diagnosis. The final model misses some AF-positive record-event groups at the selected threshold. This limitation must be reported clearly. It would be misleading to describe the system as ruling out AF. A negative PPG prediction should mean that the model did not detect AF-like evidence in the accepted PPG windows, not that AF is absent.

Repeated monitoring may reduce the practical effect of some false negatives because future windows may capture stronger evidence. However, that argument only applies after prospective longitudinal testing. The current evaluation is retrospective and record-event based. It cannot prove detection latency, patient-level detection yield over time, or alert burden in daily life.

B.4 Coverage and Abstention

SQI-based rejection creates a no-call state. Ethically, this is preferable to forcing predictions on corrupted signals, but it also creates a coverage issue. If a system abstains too often, users may believe they are being monitored when the algorithm is frequently unable to assess rhythm. A deployed system should report coverage and explain when the signal is unreliable.

Coverage should also be analysed by subgroup and context. If certain patients or conditions produce lower PPG quality, they may receive fewer valid predictions. This could create unequal screening performance. Future work should therefore measure accepted-window coverage as well as classification metrics.

B.5 Bias and Subgroup Robustness

PPG quality can vary with skin tone, perfusion, age, comorbidities, oedema, vasoconstriction, sensor placement, and movement [6, 32]. ICU-derived PPG may not represent consumer wearable PPG across all demographic groups. A model that performs well on aggregate may still underperform for particular subgroups.

The current project does not provide a complete fairness audit. It reports aggregate p00–p02 performance, but it does not stratify by demographics or clinical subgroups. This is a limitation. Before deployment, the model should be evaluated for subgroup differences in sensitivity, specificity, precision, and coverage. It should also be tested on device-specific PPG data, because signal characteristics can differ across sensors.

B.6 Data Governance

The project uses de-identified, credentialed clinical waveform data [9, 10, 30, 31, 33]. Even when data are de-identified, they remain sensitive because they originate from patient care. Responsible use requires following the data use agreement, storing data securely, reporting only aggregate results, and avoiding any attempt to re-identify patients. The final paper does not include patient-level examples that could identify individuals.

The embedded deployment work uses a holdout CSV converted into firmware test data. In a real deployment, additional governance would be required for live sensor data, including consent, retention policy, security, access control, and clear communication of what the device does and does not diagnose.

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